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Using thermochronometry and low-temperature demagnetization to accurately date Precambrian paleomagnetic poles

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Abstract. Dating magnetizations in Precambrian rocks is increasingly important in the attempt to unravel Precambrian plate configurations and supercontinent assemblages. We used low-temperature demagnetization and modern thermochronometric methods to dissect a multicomponent magnetization of the Glamorgan Gabbro, Ontario, previously studied by *Buchan and Dunlop* [1976] and *Berger and York* [1981]. We found that the Hb_A component is a primary thermoremanent magnetization carried by single-domain magnetite. The new paleomagnetic pole position (32.6°S latitude, 141.9°E longitude) is not significantly different from the published pole; however, the cooling history suggests that the 1015 ± 15 Ma magnetization age is older than previously thought (980 Ma). The new age produces a better fit in Rodinia supercontinent reconstructions. The refined Hb_B pole (29.9°N latitude, 169.9°E longitude) is carried by multidomain-type magnetite and pyrrhotite. A possible ~175°C thermal event at ~600 Ma recorded by K-feldspar could be responsible for remagnetizing the multidomain grains. The new age for Hb_B is 220 m.y. younger than the previous estimate, raising questions concerning the ages of similar poles from the Canadian Grenville Province.

1. Introduction

A major goal of Precambrian paleomagnetic studies has been to date the magnetic components typically found in ancient rocks to construct accurate apparent polar wander paths (APWP). This goal is not often achieved because it is difficult to assign ages unequivocally to discrete magnetization components. Many multicomponent magnetizations are present in slowly cooled metamorphic rocks or deep-crustal intrusions [*Buchan et al.*, 1977], where standard paleomagnetic field tests of stability (fold, baked contact, and conglomerate) are difficult or impossible. If the individual components of magnetization could accurately be tied to the thermal history, well-defined Precambrian apparent polar wander paths could result.

Dating magnetization components in multicomponent rocks has relied on single-domain (SD) fine magnetic particle theory [e.g., *Pullaiah et al.*, 1975; *Dodson and McClelland-Brown*, 1980] to relate laboratory observations of thermal demagnetization behavior to a rock's thermal history. Several studies have documented departures from this theory [e.g., *Middleton and Schmidt*, 1982; *Kent*, 1985; *Jackson and Van der Voo*, 1986]. The theoretical approaches of *Pullaiah et al.* [1975] and *Dodson and McClelland-Brown* [1980] assume that SD magnetic grains (0.02–1 μm) carry the natural remanent magnetization (NRM) of a rock. It has become increasingly apparent that many rocks contain a distribution of grain sizes ranging from small SD grains to larger multiple-domain (MD) grains, some containing domain walls that are "hard" to move or unpin (SD-type) and some which are relatively easy to unpin (MD-type). Recent studies [*Dunlop and Özdemir*, 1993; *Dunlop and Xu*, 1994; *Xu and Dunlop*, 1994] suggest that the observed departures from theory appear to arise from the presence of MD-type behavior. Theoretical and experimental studies [*Dunlop and Xu*, 1994; *Xu and Dunlop*, 1994] indicate that MD magnetite may acquire a partial thermoremanent magnetization (pTRM) over a narrow temperature range but that a broad temperature range may be required to remove it during thermal demagnetization. These observations have important consequences for interpretation of multicomponent magnetizations in rocks containing a mixture of SD-type and MD-type grains. The broad temperature ranges required to remove pTRMs in MD grains may overlap and obscure the SD signal. *Hyodo and Dunlop* [1989] have shown that linear trajectories on orthogonal projections of thermal demagnetization data can actually be the geologically meaningless vector sum of two components, one carried by SD magnetite and the other by MD magnetite. Even small MD grains with high coercivity (e.g., pseudo-single-domain grains of *Halgedahl and Fuller* [1983]) may have complicated thermal unblocking behavior [*Dunlop and Argyle*, 1991; *Halgedahl*, 1993].

MD-type effects must be removed if the unblocking temperature spectrum of a magnetization is to confidently be related to the rock's thermal history. *Dunlop and Argyle* [1991] suggested removing the MD component using low-temperature demagnetization (LTD) [*Kobayashi and Fuller*, 1968; *Merrill*, 1970]. Liquid nitrogen temperatures (77 K) are effective at unpinning domain walls in synthetic MD grains, while leaving SD-like behavior, pinned by shape anisotropy, intact. The sample can then be demagnetized thermally at high temperatures to determine its SD component of magnetization and unblocking temperature spectrum.

Buchan et al. [1977] and *Berger et al.* [1979] attempted to date paleomagnetic remanence directly by relating magnetic unblocking temperatures measured in the laboratory to the rock cooling history obtained from $^{40}Ar/^{39}Ar$ thermochronometry. *Berger et al.* [1979] applied this technique to paleomagnetic poles

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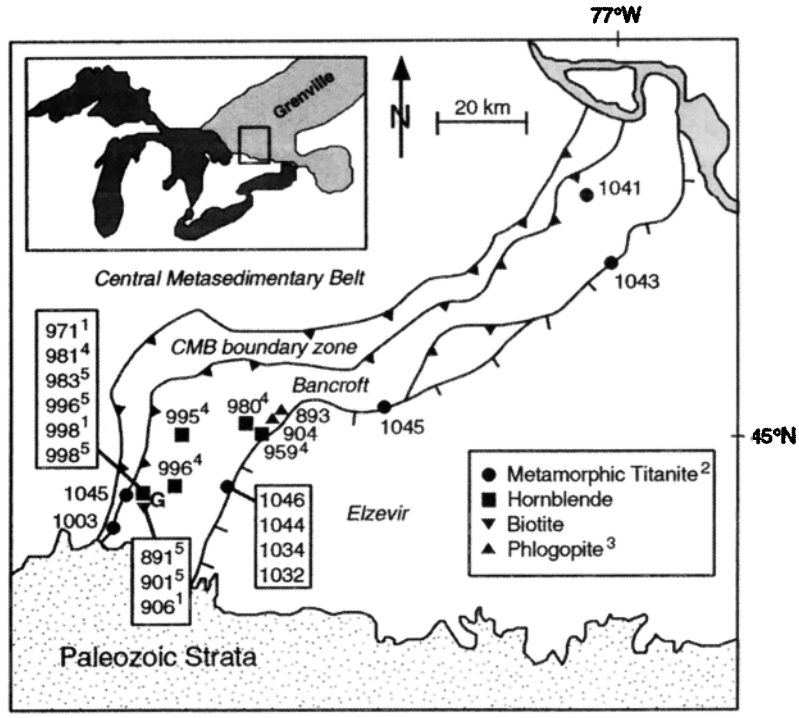


Figure 1. Schematic map of the Bancroft terrane in the Grenville province showing the location of the Glamorgan intrusive (G) and available thermochronometric cooling ages from (1) *Berger and York* [1981], (2) *Mezger et al.* [1991], (3) *Cosca et al.* [1992], (4) *Cosca et al.* [1995], and (5) this study. (Map modified from *Cosca et al.* [1995].)

(Hb_A and Hb_B) isolated from the Haliburton intrusions of the Canadian Grenville Province by *Buchan and Dunlop* [1976]. Since their study did not allow for the then poorly understood complexities of MD remanence, we have revisited the Glamorgan intrusion to test the LTD technique for isolating SD remanence. This unit is a good choice for testing the LTD method for two reasons. First, the paleomagnetic components previously isolated span and help define much of the Grenville part of the North American APWP, and second, the ages of these components have long been relied on to calibrate other Grenville poles, which lack reliable radiometric ages.

2. Geologic Setting

The Glamorgan intrusion is one of three plutons (Dudmon and Bark Lake) located within the Hastings-Haliburton Highlands of the western part of the Grenville Province of Canada (Figure 1). All three belong to the Bancroft terrane, which is interpreted to be an accreted terrane that docked with a continental margin to the northwest during the Grenville orogeny, about a billion years ago [*Dickin et al.*, 1988; *Hoffman*, 1989]. The Bancroft terrane is primarily composed of middle to upper amphibolite facies marble and metapsammites intruded by granodioritic orthogneiss, nepheline syenite, and carbonatite [*Heaman et al.*, 1987]. The Glamorgan igneous complex is one such intrusion. The part of the Glamorgan complex studied here (44°53'15"N, 78°23'15"W) is referred to as "the lobe" due to its outcrop pattern [*Downing*, 1973]. The lobe is unique in that it has escaped the penetrative effects of regional metamorphism experienced by the rest of the complex. The remarkable preservation of primary igneous textures within the gabbro has been attributed to the surrounding carbonate rocks which absorbed the bulk of the deformation [*Downing*, 1973]. Within the lobe the gabbro exhibits a range of grain sizes from coarse to relatively fine with a uniform chemical composition [*Downing*, 1973]. In addition, lamprophyre dikes ranging from tens of centimeters to several meters in width are present. These dikes have been interpreted as predating (1041 Ma) the Grenville Orogeny based on field relations [*Downing*, 1973].

Buchan and Dunlop [1976] isolated three magnetic components (Hb_A, Hb_B, and Hb_C) from the Haliburton intrusions. The Hb_A component (36°S, 142.5°E) was interpreted as the component with highest blocking temperature in the Glamorgan Gabbro. Hb_A is carried by magnetite, and *Berger et al.* [1979] assigned it an age of 980 Ma by comparing its unblocking temperature, corrected for magnetic relaxation, to a thermochronometric cooling curve for the region. The Hb_B component (24.5°N, 172.3°E) appears to have a lower and broader spectrum of unblocking temperatures than the Hb_A component and was interpreted as reflecting a low-temperature pTRM carried by magnetite with an age of ≤820 Ma derived in the same manner. Two samples exhibited stable end points that persisted at demagnetization temperatures above 600°C. *Buchan and Dunlop* [1976] interpreted this as a chemical remanent magnetization residing in hematite. With the exception of these two samples, all of the Hb_B directions were obtained by vector subtraction. Consequently, the Hb_B component shows more directional dispersion than Hb_A. The Hb_C component (3°S, 167°E) appears to be carried by magnetite and only occurs in a localized region of the Glamorgan unit. Hb_C was assigned an intermediate age of 900 Ma based on its intermediate pole position. It is unique in that it displays both low (~350°C) and high (~580°C) unblocking temperatures [*Buchan and Dunlop*, 1976].

Igneous titanite U/Pb ages from the Bancroft terrane range from 1074 to 1041 Ma and indicate that the majority of igneous activity preceded peak metamorphism at 1041 Ma [*Mezger et al.*,

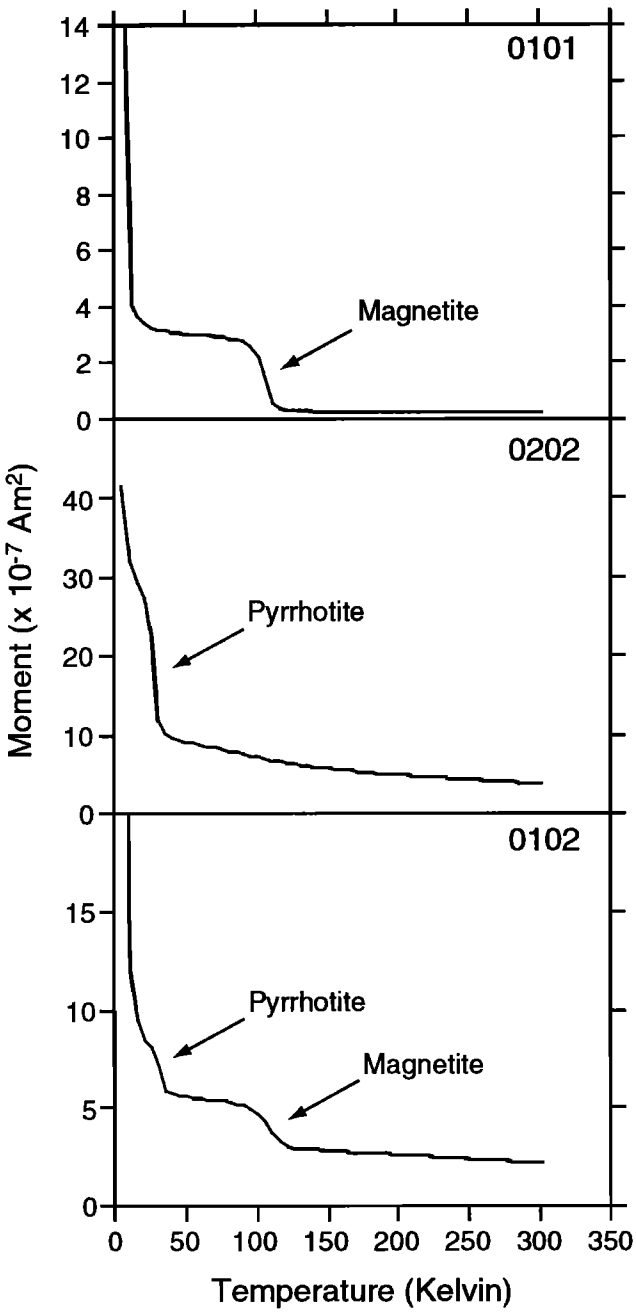


Figure 2. Saturation remanence as a function of temperature. Steps in the remanence curve represent crystallographic reorganization. Note that there are two steps, one at ~34 K and the other at ~120 K. These coincide with a transition in pyrrhotite and the Verwey transition in magnetite, respectively. Note that pyrrhotite and magnetite occur both separately and together.

1991]. Hornblende $^{40}Ar/^{39}Ar$ ages indicate that following the Grenville orogeny, the Bancroft terrane cooled through ~500°C uniformly on a regional scale (Figure 1). Although *Cosca et al.* [1995] call for 9° to 15° of post-Grenville horizontal axis tilt of the Bancroft terrane, their argument is based largely on one anomalously low hornblende $^{40}Ar/^{39}Ar$ age near the Bancroft shear zone and will not be assumed, in this manuscript, to have affected the magnetization. *Berger and York* [1981] constructed a cooling history for the intrusives of the Haliburton Highlands. Their Glamorgan hornblende and biotite $^{40}Ar/^{39}Ar$ data, used together with currently accepted closure temperatures [*McDougall and Harrison*, 1988], has been supplemented with our new hornblende, biotite, and K-feldspar $^{40}Ar/^{39}Ar$ data from the lobe.

3. Paleomagnetism

3.1. Methods

Twelve oriented hand samples were collected as pilot samples from three relatively fresh road cuts near Glamorgan sites G3 and G4 of the *Buchan and Dunlop* [1976] study. The samples were oriented with a magnetic compass, and the orientations were

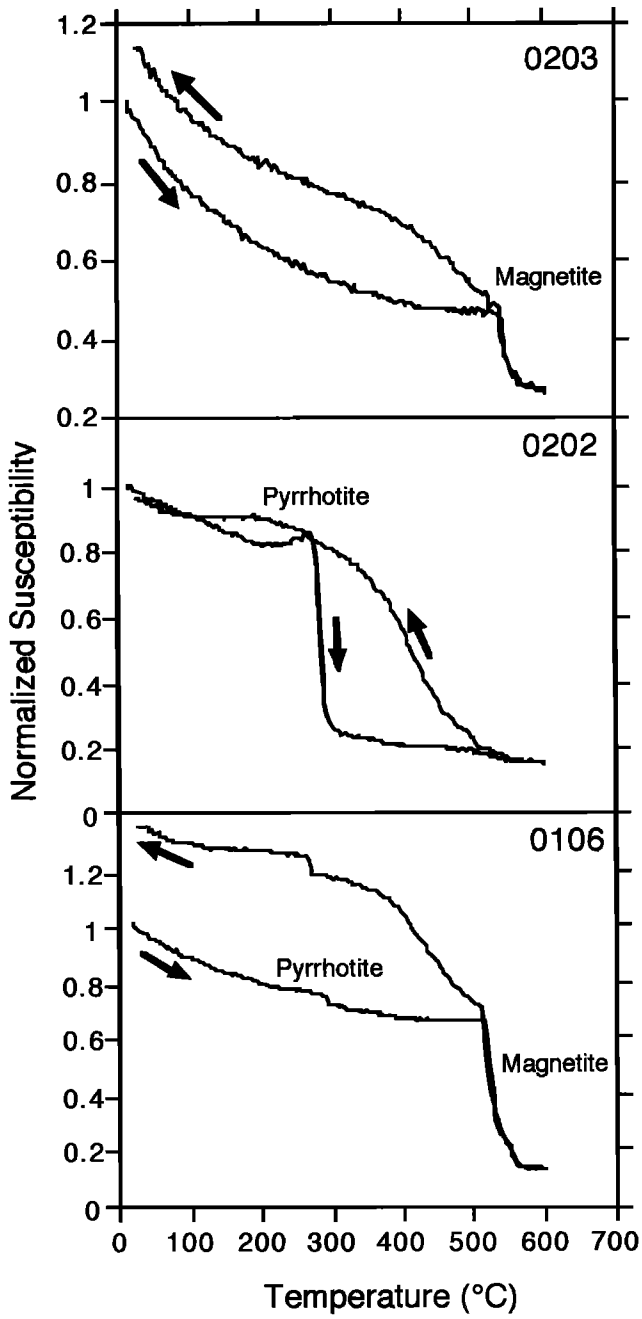


Figure 3. Curie point analysis by susceptibility versus temperature. Note that steps in the heating curve near ~300°C and 570°C represent the Curie points for pyrrhotite and magnetite, respectively. Note that pyrrhotite and magnetite occur both separately and together.

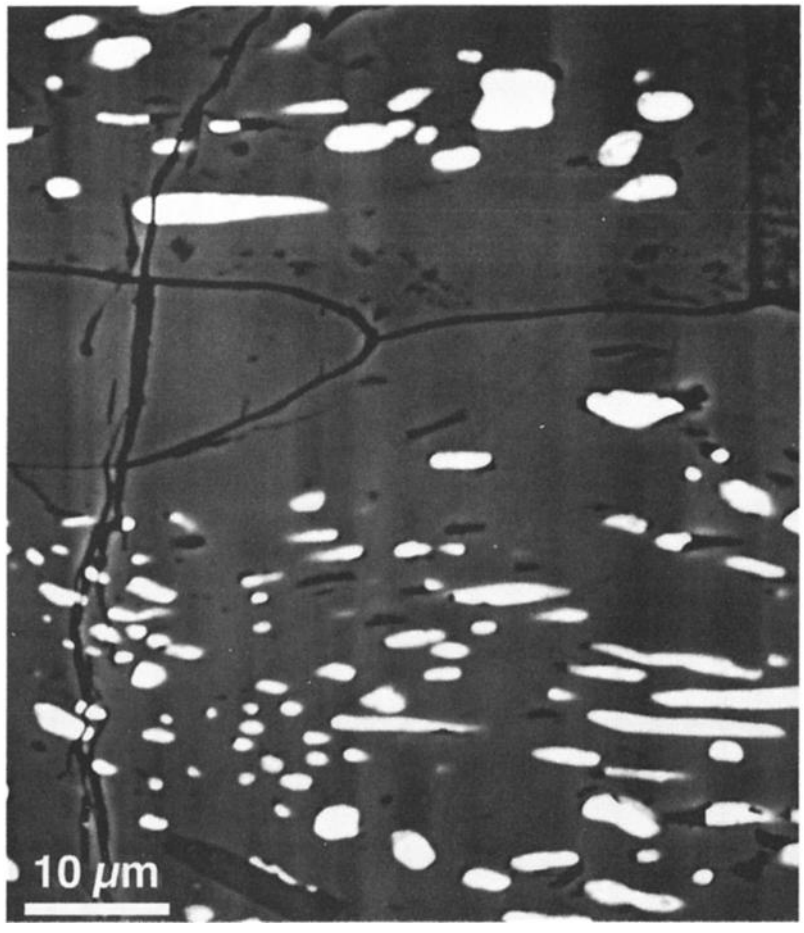


Figure 4. Back scattered electron photomicrograph showing symplectic growth of magnetite (white) in orthopyroxene (dark) host. Crystallographic lineations of magnetite are randomly oriented within single orthopyroxene host grains. This crystal texture is consistent with symplectic growth at high temperature. Note that magnetite grains range in size from ~0.5 μm to ~10 μm.

checked with a second compass held far enough away from the outcrop to avoid errors associated with outcrop magnetic fields. Several standard paleomagnetic core samples were drilled from these hand samples in the laboratory for both paleomagnetic and rock magnetic analysis. These pilot samples were later supplemented with 40 cores from the same outcrops. These samples were oriented in situ with a sun compass verified with a magnetic compass. Orientations determined by sun and magnetic compass agreed to within 0.5°. Each core yielded two specimens. One was analyzed by standard thermal demagnetization, and the other was thermally demagnetized following LTD.

Remanence measurements were made on a CTF Systems Inc., two-axis, superconducting magnetometer at Lehigh University. Two suites of samples were analyzed: one by standard thermal demagnetization (12–14 demagnetization steps up to 580°C) to isolate the magnetic components originally observed by *Buchan and Dunlop* [1976] and the other by detailed thermal demagnetization after LTD. LTD treatment was carried out in a shallow, wide-mouthed dewar inside a horizontal double layer μ-metal shield. The ambient field magnitude within the dewar was <5 nT. The dewar was filled with liquid nitrogen with the aid of a custom-made funnel. As the liquid nitrogen boiled off, the samples remained fully submerged for ~30 min. The samples were allowed to return to room temperature over the course of ~1 hour prior to the next low-temperature cycle. A total of five cycles were used to ensure complete demagnetization of the MD grains, as suggested by *Borradaile* [1994].

Detailed thermal demagnetization of the samples was conducted on a Schonstedt TSD-1 thermal demagnetizer. To ensure accurate thermal demagnetization, the demagnetizer was calibrated by monitoring the temperature at the center of 2.5-cm-diameter specimens with a K-type thermocouple. Several sample positions within the sample boat were monitored in this manner, allowing the creation of a set of temperature-time reference curves for each thermal demagnetization step. These curves allow the temperature for each sample position to be determined for each demagnetization step. Errors in temperatures are ±5°C. Before initial heating, and after each subsequent demagnetization step, magnetic susceptibility was measured on a SI-2 Sapphire Instruments susceptometer to detect thermally induced changes in magnetic mineralogy. Characteristic remanent magnetizations were determined using principal component analysis [*Kirschvink*, 1980]. The mean paleomagnetic pole was determined by averaging the individual sample directions [*Fisher*, 1953]. No structural correction was applied to the data.

Rock magnetic experiments were conducted both at Lehigh University and the Institute for Rock Magnetism at the University of Minnesota to characterize the magnetic minerals in the rocks.

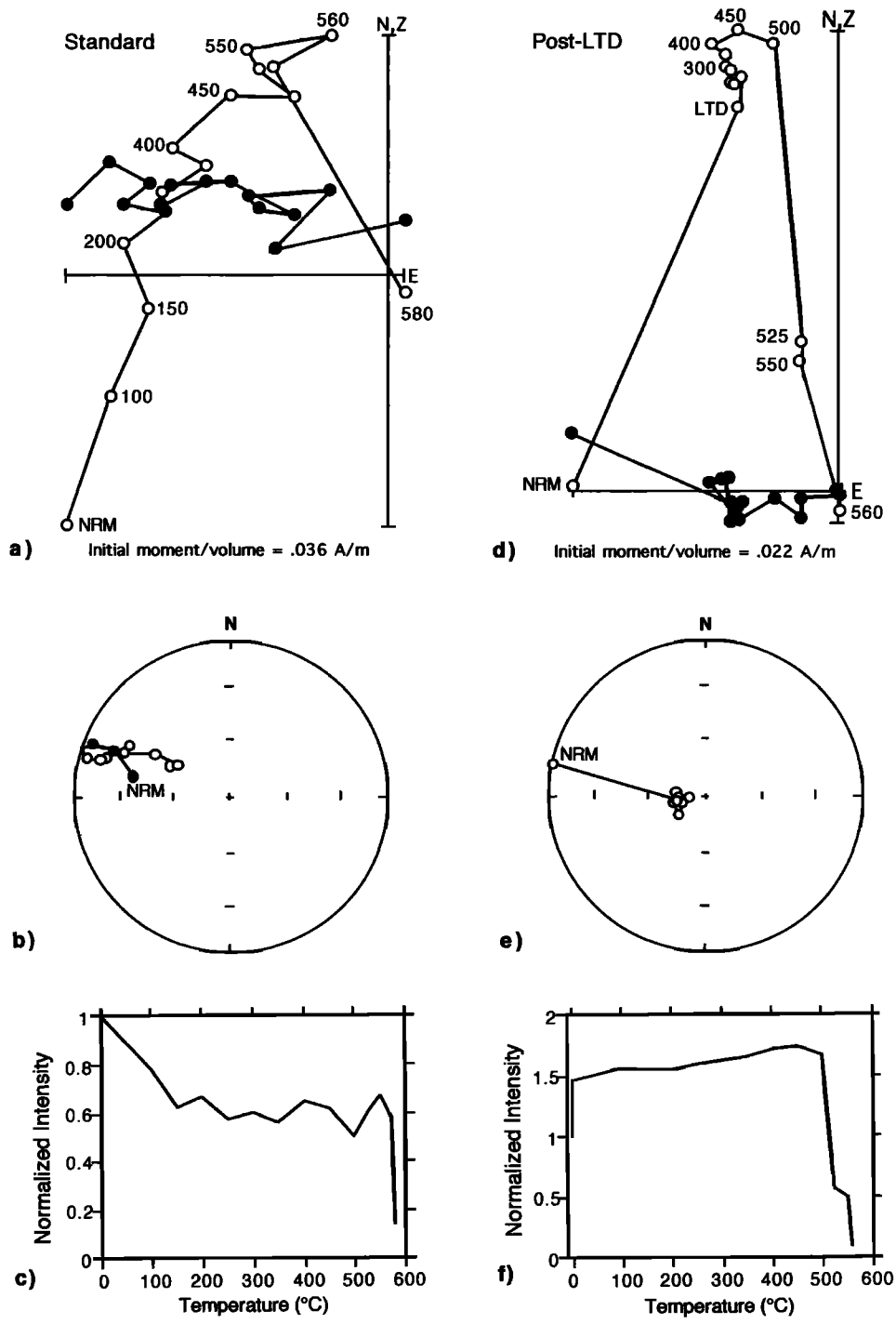


Figure 5. Results of thermal demagnetization experiments both (a-c) before and (d-f) after LTD for a sample containing only magnetite. Figures 5a and 5d are orthogonal projections of demagnetization path. Open symbols are inclination projected onto a vertical plane; solid symbols are declination. Figures 5b and 5e are equal-area projections of thermal demagnetization data. Open symbols are negative inclinations. Figures 5c and 5f show normalized intensity as a function of demagnetization temperature.

In order to understand the relative magnetic coercivities of the magnetic minerals present, two untreated specimens taken from each of the 12 pilot hand samples were given three isothermal remanent (IRM) components (1.2, 0.4, and 0.15 T) oriented orthogonally [*Lowrie*, 1990]. One suite was treated with standard thermal demagnetization; the other suite was thermally demagnetized after LTD.

At the Institute for Rock Magnetism, low-temperature crystallographic transitions were identified by measuring a saturation remanence acquired in a 2.5 T field while warming from 5 K to 300 K on a Quantum Design MPMS1 superconducting susceptometer.

Curie points of the magnetic minerals were determined for 10 of the 12 pilot samples by measuring low-field susceptibility as a

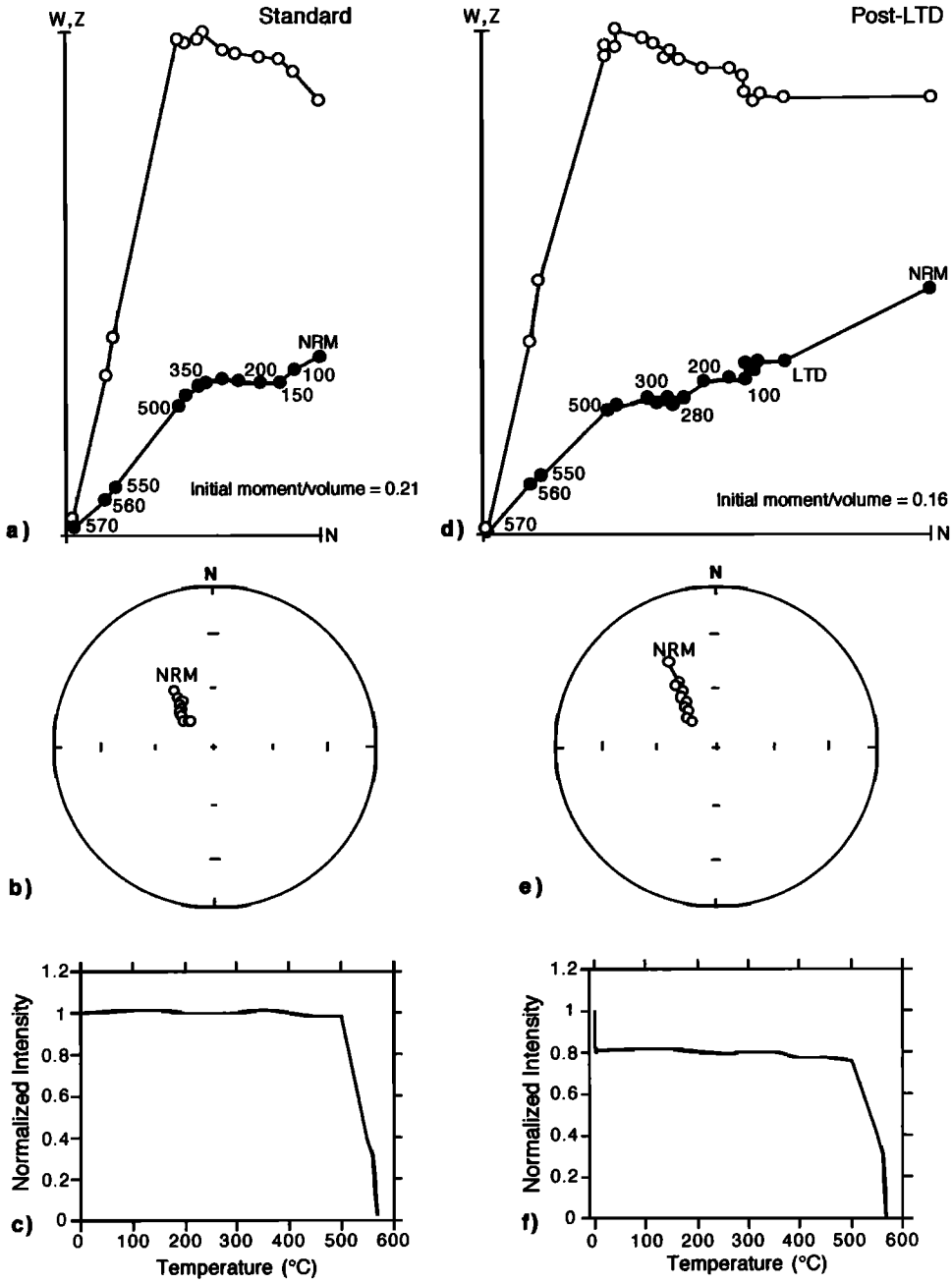


Figure 6. Results of thermal demagnetization experiments both (a-c) before and (d-f) after LTD for a sample containing both magnetite and pyrrhotite. Figures 6a and 6d are orthogonal projections of demagnetization path. Open symbols are inclination projected onto a vertical plane; solid symbols are declination. Figures 6b and 6e are equal-area projections of thermal demagnetization data. Open symbols are negative inclinations. Figures 6c and 6f show normalized intensity as a function of demagnetization temperature.

function of temperature on a Geofyzika Kappabridge susceptometer. The samples were heated in a constant stream of argon gas to reduce the affects of oxidation.

Highly polished thin sections were examined with both a reflected light microscope and a JEOL 733 electron microprobe.

3.2. Results

The 0.4 and 1.2 T IRM components were more resistant to thermal demagnetization than the 0.15 T component, and the 0.15 T component was the most susceptible to removal by LTD. This is consistent with low-temperature unpinning of the low-coercivity, MD-type component. The resistance to thermal demagnetization of the high-coercivity component suggests that SD grains are also present.

Low-temperature transitions at 34 K (pyrrhotite) and 120 K (Verwey transition in magnetite) were observed (Figure 2). On the basis of these observations the samples can be divided into three different groups of approximately equal size according to their magnetic mineralogies. One group contained pyrrhotite as the only magnetic phase, another contained just magnetite, and the third contained both pyrrhotite and magnetite (Figure 2). This is consistent with the unblocking behavior observed.

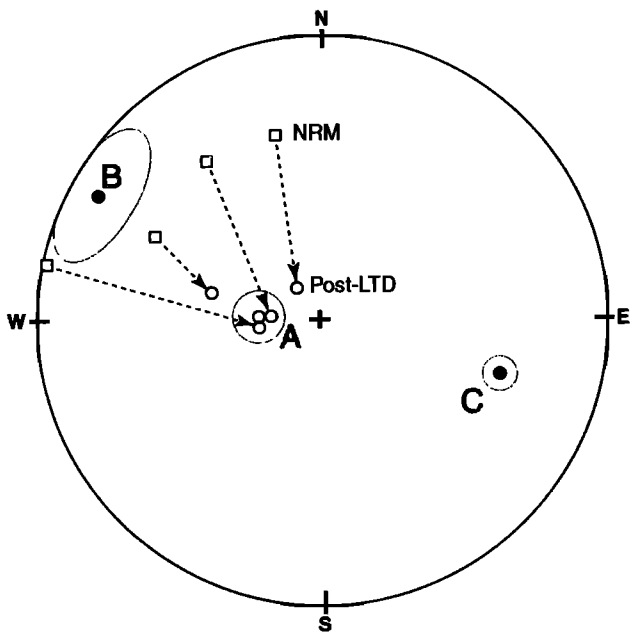


Figure 7. Equal-area projection of low-temperature demagnetization results for samples that contain magnetite as the only magnetic phase. Note the dramatic removal of the B component overprint (squares). Open symbols are upward directions. The Hb_A, Hb_B, and Hb_C directions of *Buchan and Dunlop* [1976] are plotted for reference.

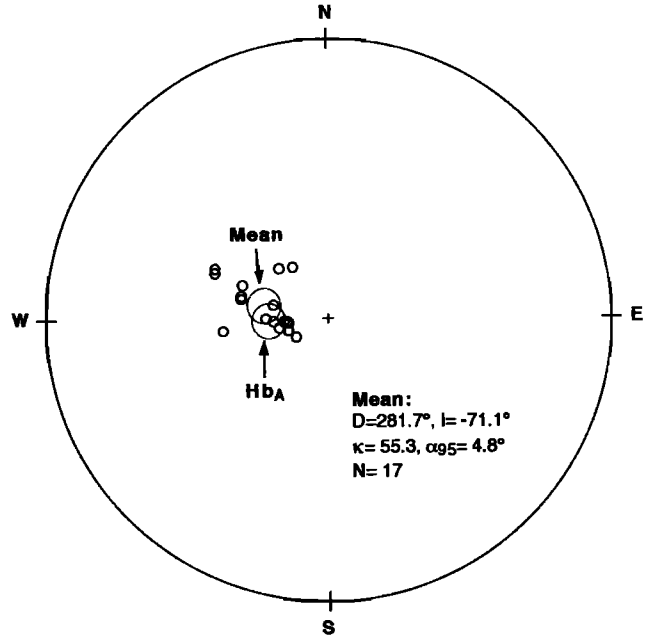


Figure 8. Equal-area projection of post-LTD sample directions that exhibit the Hb_A component. Note that the mean direction is not significantly different from the Hb_A component of *Buchan and Dunlop* [1976]. Open symbols indicate upward directions. Circles are α_{95}.

In addition to the two low-temperature transitions found, two Curie points were observed; one at ~320°C and the other at 570°C (Figure 3). The ~320°C Curie point is due to pyrrhotite; the 570°C Curie point is due to magnetite. The Curie point curves were irreversible. This irreversibility of the curves suggests that pyrrhotite is oxidized to magnetite during heating, consistent with a rapid rise in susceptibility above 500°C observed during the thermal demagnetization experiments.

Optical examination of thin-sections from each of the 12 pilot samples revealed that the coarse-grained gabbro has a different magnetic mineralogy from the fine-grained gabbro. The coarse-grained gabbro contains minor amounts of fine-grained (0.5–10

Table 1. Summary of Hb_A Component Data

Buchan and Dunlop [1976] Suite					LTD Suite From This Study				
Site	*N*	*D*	*I*	α_{95}	Sample	*N*	*D*	*I*	MAD*
B1	5	233.7	-74.6	12.7	1010201L	4	283.9	-64.0	7.2
B2	2	267.2	-57.2	N/A	1020201L	6	316.1	-70.1	0.7
B5(A)	1	308.1	-64.2	N/A	1030201L	4	292.7	-54.5	5.5
B8	1	288.0	-66.1	N/A	2030101L	7	239.4	-79.4	1.3
B9	1	227.7	-75.0	N/A	96A02L	5	269.7	-72.1	7.4
B12	5	212.4	-71.7	5.9	96A03L	5	263.4	-59.3	8.4
B14	3	269.5	-62.3	16.4	96A05L	7	283.4	-73.9	3.8
D1	3	253.7	-67.6	9.7	96A06L	6	259.0	-75.6	2.2
D2	3	279.8	-69.9	13.8	96A07L	8	254.6	-78.6	3.2
D4	1	256.9	-72.5	N/A	96A08L	6	265.8	-77.4	3.8
D5	4	165.6	-83.3	19.4	96C05L	4	293.9	-53.8	3.1
D6	4	266.0	-72.0	6.3	96C06L	5	284.4	-64.1	5.1
D11	2	288.7	-68.9	N/A	96C07L	7	291.6	-63.6	1.1
G3	3	344.1	-66.2	12.0	96C08L	6	262.1	-78.9	3.5
G4	4	293.2	-67.4	8.0	96C09L	6	268.2	-78.2	1.5
					96C10L	4	268.2	-73.8	3.2
					96D02L	6	327.4	-71.8	10.9
Mean†	42	265.2	-74.1	4.3		17	281.7	-71.1	α_{95}=4.8
Combined mean						59	277.5	-71.8	α_{95}=3.8

*Maximum angular deviation [*Kirschvink*, 1980].
†Unit weight to samples.

Table 2. Summary of Hb_B Component Data

Buchan and Dunlop [1976] Suite					Suite From This Study				
Sample	*N*	*D*	*I*	α_{95}	Sample	*N*	*D*	*I*	MAD*
149	3	315.0	16.1	12.9	01010101S	3	318.1	32.2	0.2
138	3	271.0	-3.7	18.7	01020201L	10	343.4	21.9	8.2
139	3	319.0	-10.4	7.8	01030101S	7	306.6	-12.5	2.3
409	3	304.7	13.8	10.0	96C02S	5	300.4	18.9	10.3
160	3	318.6	10.7	21.9	96C05L	6	303.7	17.7	10.5
420	3	295.1	6.6	9.7	96C10S	7	326.7	43.2	15.4
421	3	285.2	31.6	8.7					
302	3	262.1	-1.0	22.7					
184	3	328.8	47.7	9.0					
186	3	306.3	-13.7	19.2					
857	3	304.0	8.0	19.6					
Mean†	11	299.9	10.3	15.7		6	315.7	21.1	α_{95} = 20.7
Combined Mean						17	305.7	14.1	α_{95} = 12.3

*Maximum angular deviation [*Kirschvink*, 1980].
†Unit weight to samples.

μm) opaque minerals hosted by orthopyroxene. Quantitative microprobe analyses verified that these opaques are pure iron oxide. They are assumed to be magnetite based on the Curie point data, and their grain sizes range from SD to MD due to their elongated crystal shape (Figure 4). The magnetite grains exhibit a symplectic crystal texture [*Barton and Van Gaans*, 1988; *Barker*, 1990] (Figure 4). Consequently, the elongated crystals are randomly oriented both within and between grains of orthopyroxene. Macroscopic multidomain (>1.6 μm [*Soffel*, 1977]) crystals of pyrrhotite exhibiting primary igneous textures are also present in varying concentrations. No veins were observed in the coarse-grained samples. Fine-grained gabbros, however, do exhibit veins containing fine-grained sulfide crystals. The vein sulfides have not been confirmed to be pyrrhotite but are assumed to be pyrrhotite based on the Curie point data. Symplectic magnetite was not observed in the fine-grained samples.

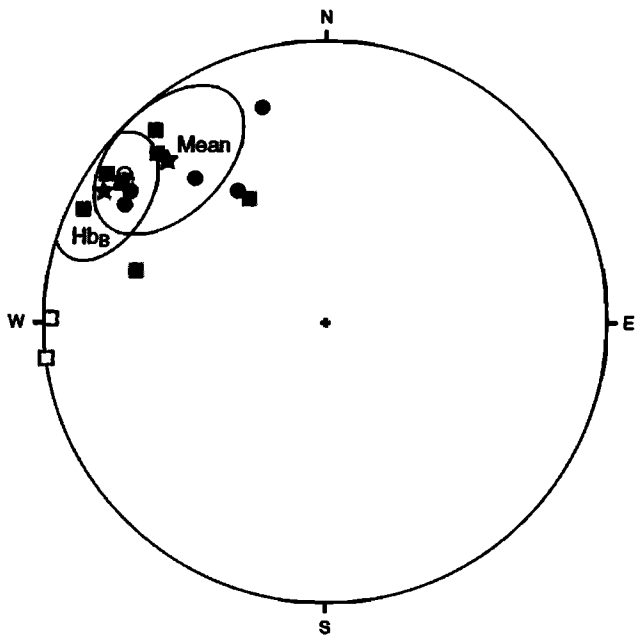


Figure 9. Equal-area projection of Hb_B component directions isolated by standard thermal demagnetization (black symbols) compared to the Hb_B component of *Buchan and Dunlop* [1976] (gray symbols). Open symbols indicate upward directions. Circles are α_{95}.

Directions similar to two of the three components isolated by *Buchan and Dunlop* [1976] have been identified in the rocks studied here. Directions similar to the Hb_A component are found in samples which contain magnetite (Figures 5 and 6). In the samples that contain only magnetite, LTD successfully removes overprints completely (Figures 5 and 7). In all cases the Hb_A component unblocks over a very narrow temperature range between ~555° and ~565°C. The mean direction for 17 samples (Table 1 and Figure 8) from the LTD suite has a mean declination of 281.7° and inclination of -71.1° (α_{95} 4.8°) and is not statistically [*McFadden and Lowes*, 1981] distinguishable, at the 95% confidence level, from the component isolated by standard demagnetization for the Hb_A component of *Buchan and Dunlop* [1976] (*D* 272.2°, *I* -72.5°, α_{95} 6.3°). Combining our data with the Hb_A component of *Buchan and Dunlop* [1976] yields a *D* 277.5°, *I* -71.8°, α_{95} 3.8° (32.6°S, 141.9°E).

We were able to resolve the Hb_B component in six samples (Table 2). In all cases, Hb_B was superimposed on the Hb_A component (Figure 6). The mean direction (*D* 315.7°, *I* 21.1°, α_{95} 20.7°) is not significantly different from the mean given by *Buchan and Dunlop* [1976] (*D* 299.9°, *I* 10.3°, α_{95} 15.7°), but once combined with *Buchan and Dunlop's* [1976] results in a slightly more well-defined direction of *D* 305.7°, *I* 14.1°, α_{95} 12.3° (29.9°N, 169.9°E) (Figure 9). Samples that contain pyrrhotite and magnetite typically show both the Hb_A and Hb_B components. The Hb_B component defines a linear demagnetization path below ~320°C, whereas the Hb_A component is stable above 320°C (Figure 6). Samples that contain only magnetite exhibit a dramatic jump from a Hb_B component NRM overprint direction to a stable Hb_A component direction immediately after LTD suggesting that MD magnetite is also a carrier of the Hb_B component (Figures 5 and 7). There is no clear evidence of the Hb_B component in the samples that contain pyrrhotite as the sole magnetic phase.

Buchan and Dunlop [1976] found the Hb_C component (*D* 105.4°, *I* 34.7°, α_{95} 6.6°) in a very localized area within the Gla-

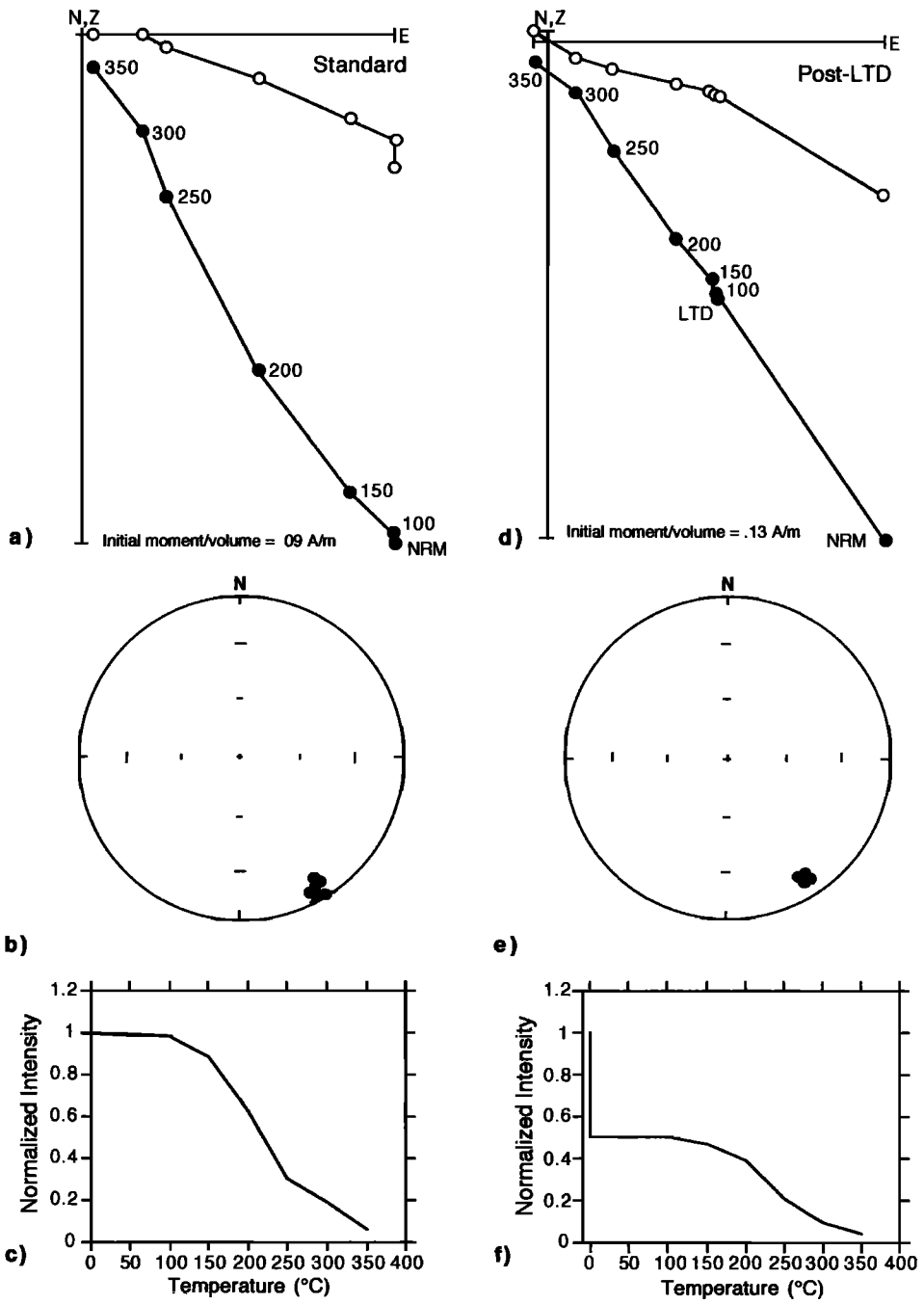


Figure 10. Results of thermal demagnetization experiments both (a-c) before and (d-f) after LTD for a sample containing only pyrrhotite. Figures 10a and 10d are orthogonal projections of demagnetization path. Open symbols are inclination projected onto a vertical plane; solid symbols are declination. Figures 10b and 10c are equal-area projections of thermal demagnetization data. Open symbols are negative inclinations. Figures 10c and 10f show normalized intensity as a function of demagnetization temperature.

morgan Igneous Complex and in several different rock types. They interpreted it as being carried by magnetite having a range of unblocking temperatures. We found a direction similar to the Hb_C component in one pyrrhotite-bearing sample. However, we also found 13 other samples that recorded well-defined pyrrhotite directions (Figures 10 and 11). These directions are widely scattered and statistically random [*Watson*, 1956]. The nature of the Hb_C component, therefore, remains uncertain.

4. Thermochronometry

4.1. Methods

Two hornblende and two biotite samples were collected directly from the same sites as the paleomagnetic samples. An additional hornblende was collected from an outcrop 1 km to the north-northeast of the paleomagnetic sampling site. K-feldspar was collected from a syenite 2 km to the south.

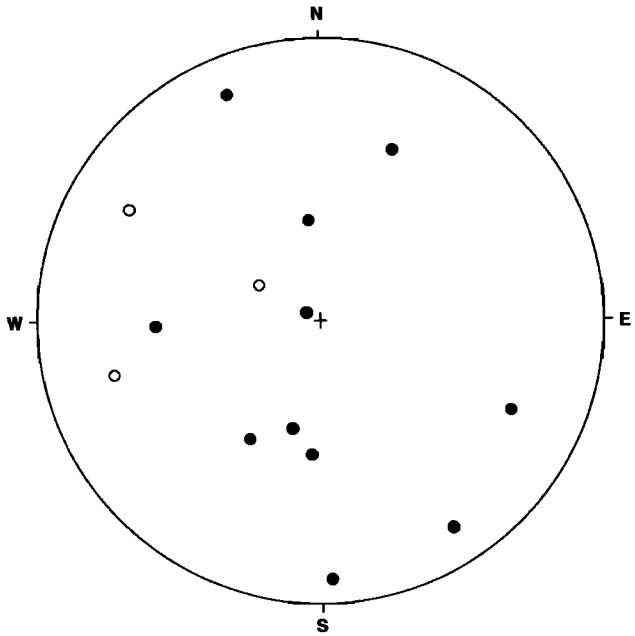


Figure 11. Equal-area projection of pyrrhotite sample directions. Note that the distribution is statistically random [*Watson*, 1956]. Open symbols indicate upward directions.

All samples were crushed and sieved to between 150 and 180 μm. Minerals were separated using heavy liquids, magnetic susceptibility, and hand-picking under a binocular microscope. Samples showed no signs of contamination or alteration. Samples (1–5 mg) were enclosed in tin packets and sealed under vacuum in quartz glass vials for irradiation in the L67 position at the University of Michigan Ford reactor. Samples were irradiated for 40 hours. MMhb-1 (520.4 ± 1.7 Ma) [*Samson and Alexander*, 1987] was used to monitor the fast neutron flux. CaF_2 and K_2SO_4 were used to quantify interfering reactions on calcium and potassium.

Argon gas was extracted from the samples in a standard double-vacuum resistance furnace. The samples were heated in a molybdenum-lined tantalum crucible with temperature control precise and accurate to 0.5%. Biotites were step heated with ~12 steps, and hornblendes were step heated with ~18 steps. The K-feldspar analysis was run using a modern isothermal-duplicate heating schedule [*Harrison et al.*, 1993] having 32 steps. The gas released on heating was subsequently cleaned to remove active gases using two SAES GP-50 titanium getters, one operated at 400°C and the other at room temperature. Temperature-dependent extraction line blanks were well characterized and never more than 5% of the ^{40}Ar beam. Argon analyses were carried out on a VG 3600 mass spectrometer equipped with an electron multiplier collector with an effective sensitivity of 2 x 10^{-16} mol/mV. Corrections for mass discrimination, decay of ^{37}Ar and ^{39}Ar, interfering reactions on calcium and potassium during irradiation, and extraction line blanks were applied to the data. Decay constants and isotopic abundances were taken from *Steiger and Jäger* [1977]. All uncertainties are reported as 1σ. Feldspar step-heating data were run through a forward diffusion model [*Lovera et al.*, 1989, 1991] to determine the continuous thermal history between ~350°C and 125°C.

4.2. Results

None of the samples show signs of significant excess argon (Tables 3-8 and Figures 12-14). Two of the hornblendes (94GG01a and Tamarack) show ~100 Ma age gradients in their age spectra over the first 10 to 20% of ^{39}Ar released (Figure 13) which may represent loss due to weathering. It is unlikely that the loss is due to thermally activated diffusion due to the lack of such loss in sample 94GG01b. Ages and errors for the two samples that showed age gradients were derived by integrating relative contributions of the last 10 to 11 steps that define a "plateau." The resulting ages of 983 ± 8 Ma (94GG01a) and 996 ± 8 Ma (Tamarack) are statistically indistinguishable from the total gas age of sample 94GG01b (998 ± 8 Ma), as well as the majority of the hornblende results from the Bancroft terrane (Table 9 and Figure 1). Biotite total gas ages of 901 ± 7 Ma and 891 ± 7 Ma

Table 3. $^{40}Ar/^{39}Ar$ Analytical Results: 94GG01a Hornblende

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
800	10	402.39	0.3	0.4812	0.6	9.7306	0.9	1.03740	1.0	1.79e-15	0.3	0.010	23.8	96.887	879.03	23.01
900	10	104.60	0.3	0.1539	0.6	0.6636	0.9	0.00498	8.6	3.96e-15	0.3	0.033	98.3	103.190	923.71	7.51
930	10	102.21	0.3	0.1575	0.6	0.4430	3.9	0.00257	26.4	2.49e-15	0.3	0.046	98.8	101.480	911.70	7.54
950	10	102.33	0.3	0.2056	0.6	0.5474	4.6	0.00204	46.1	1.94e-15	0.3	0.058	98.8	101.760	913.72	7.67
970	10	107.65	0.3	0.3828	0.6	1.0831	1.2	0.00147	45.2	2.50e-15	0.3	0.072	99.1	107.320	952.47	7.77
990	10	110.92	0.3	0.6101	0.6	1.9910	1.7	0.00207	16.9	5.06e-15	0.3	0.100	99.2	110.530	974.45	7.81
1010	10	111.89	0.3	0.6808	0.6	2.2227	0.9	0.00184	11.6	9.01e-15	0.3	0.151	99.4	111.600	981.69	7.86
1030	10	111.98	0.3	0.7030	0.6	2.2635	0.8	0.00173	9.1	1.38e-14	0.3	0.229	99.5	111.730	982.58	7.84
1050	10	111.54	0.3	0.6950	0.6	2.3479	0.8	0.00110	13.5	1.56e-14	0.3	0.317	99.6	111.480	980.89	7.84
1070	10	111.82	0.3	0.6881	0.6	2.3179	0.8	0.00113	13.0	1.49e-14	0.3	0.402	99.6	111.750	982.75	7.84
1090	10	112.61	0.3	0.6755	0.6	2.2914	0.8	0.00111	11.2	1.89e-14	0.3	0.508	99.6	112.550	988.13	7.87
1110	10	111.40	0.3	0.6521	0.6	2.1679	1.0	0.00201	6.4	2.24e-14	0.3	0.635	99.4	111.050	978.01	7.82
1130	10	112.05	0.3	0.6266	0.6	2.1361	0.8	0.00105	10.9	1.89e-14	0.3	0.742	99.6	111.990	984.34	7.89
1150	10	111.99	0.3	0.6162	0.6	2.2830	0.8	0.00095	21.2	1.36e-14	0.3	0.819	99.6	111.970	984.25	7.92
1200	10	112.46	0.3	0.6496	0.6	2.4451	0.8	0.00091	15.0	1.83e-14	0.3	0.922	99.7	112.480	987.65	7.88
1250	10	112.29	0.3	0.6378	0.6	2.4125	0.8	0.00067	60.9	7.32e-15	0.3	0.964	99.5	112.370	986.92	7.90
1350	10	111.88	0.3	0.6291	0.6	2.4261	0.8	0.00136	38.2	6.46e-15	0.3	1.000	99.1	111.760	982.82	7.93

J = (6.48 ± 0.06) x10-3; $(^{36}Ar/^{37}Ar)_{Ca}$ = (6.91 ± 0.69) x10-5; $(^{39}Ar/^{37}Ar)_{Ca}$ = (9.42 ± 0.94) x10-4; $(^{40}Ar/^{39}Ar)_K$ = (2.62 ± 0.26) x10-2; $(^{38}Ar/^{39}Ar)_K$ = (1.160 ± 0.12) x10-2; mass discrimination is 1.0245 ± 0.01; weight is 0.0052 mg. Dur., duration; s.d., standard deviation.

*In percent.

Table 4. $^{40}Ar/^{39}Ar$ Analytical Results: 94GG01b Hornblende

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
600	10	6071.30	1.2	4.9281	1.3	7.7822	3.7	20.32600	1.6	9.08e-17	1.2	0.001	1.1	65.556	638.70	488.27
700	10	222.35	3.5	0.5661	4.0	21.7900	3.8	0.40926	4.1	1.02e-16	3.5	0.003	43.3	103.970	929.18	33.19
800	10	122.98	0.5	0.4067	0.7	9.9277	3.2	0 04297	9.9	3.36e-16	0.4	0.007	86.7	111.510	981.08	12.22
880	10	123.00	0.5	0.3622	1.0	1.9390	3.5	0.02475	12.8	5.00e-16	0.5	0.014	91.7	115.910	1010.70	10.83
930	10	113.38	0.6	0.3828	0.8	1.7490	3.4	0.00792	17.8	5.54e-16	0.6	0.021	95.3	111.230	979.19	9.23
950	10	113.76	0.5	0.4257	0.7	1.7342	2.8	0.00312	69 2	3.87e-16	0.4	0.026	95.4	113.030	991.40	9.47
990	10	115.51	0.4	0.6294	0.8	2.5640	1.6	0.00512	20.6	7.51e-16	0.4	0.036	96.7	114.300	999.94	8.47
1030	10	114.45	0.3	0.8598	0.6	3.6891	0.9	0.00239	11.3	2.87e-15	0.3	0.075	98.8	114.190	999.18	7.98
1060	10	112.74	0.3	0.8260	0.6	4.1304	0.8	0.00232	4.7	1.00e-14	0.3	0.208	99.3	112.550	988.13	7.88
1070	10	111.67	0.3	0.7790	0.6	4.1701	0.8	0.00309	5.3	6.00e-15	0.3	0.289	98.9	111.250	979.37	7.83
1085	10	112.99	0.3	0.7698	0.6	4.2193	0.8	0.00165	8.7	6.12e-15	0.3	0.371	99.3	113.010	991.26	7.90
1100	10	114.05	0.3	0.7835	0.6	4.1539	0.9	0.00178	6.8	6.53e-15	0.3	0.458	99.3	114.030	998.10	7.95
1120	10	114.20	0.3	0.7845	0.6	4.2306	0.8	0.00223	9.2	6.20e-15	0.3	0.541	99.2	114.060	998.31	7.94
1140	10	114.51	0.3	0.8058	0.6	4.3623	0.8	-0.00090	100.0	5.17e-15	0.3	0.610	99.9	115.320	1006.80	8.00
1180	10	115.39	0.3	0.8451	0.6	4.5429	0.9	0.00197	4.7	1.27e-14	0.3	0.779	99.4	115.360	1007.10	8.00
1210	10	115.49	0.3	0.8345	0.6	4.4695	0.8	0.00124	16.8	5.24e-15	0.3	0.849	99.4	115.670	1009.10	8.02
1250	10	115.48	0.3	0.8123	0.6	4.4003	0.8	0.00171	9.4	6.43e-15	0.3	0.935	99.3	115.520	1008.10	8.01
1350	10	115.38	0.3	0.7937	0.6	4.3953	0.9	0.00203	11.1	4.85e-15	0.3	1.000	99.0	115.320	1006.80	8.00

$J = (6.48 \pm 0.06)$ x10-3; $(^{36}Ar/^{37}Ar)_{Ca} = (6.91 \pm 0.69)$ x10-5; $(^{39}Ar/^{37}Ar)_{Ca} = (9.42 \pm 0.94)$ x10-4; $(^{40}Ar/^{39}Ar)_K = (2.62 \pm 0.26)$ x10-2; $(^{38}Ar/^{39}Ar)_K = (1.160 \pm 0.12)$ x10-2; mass discrimination is 1.0288 ± 0.01; weight is 0.0046 mg. Dur., duration; s.d., standard deviation.

*In percent.

agree with sample G4-22 (906 ± 6 Ma) of *Berger and York* [1981]. Modeling of the K-feldspar diffusion domains reveals that its age spectrum can be described by two end-member models: linear slow cooling between 275°C and 130°C from ~850 to 650 Ma or slow cooling followed by a ~1 m.y. thermal pulse of ~175°C at ~600 Ma (Figure 14d). Although we favor a relatively short pulse of ~175°C, models incorporating longer, lower temperature pulses are also viable. Using currently accepted closure temperatures for hornblende (500° ± 50°C) and biotite (300° ± 50°C) [*McDougall and Harrison*, 1988], the combination of the hornblende and biotite data from *Berger and York* [1981], this study, and the mean metamorphic titanite U/Pb age from *Mezger et al.* [1991] yield a Grenville cooling history for the Glamorgan gabbro (Table 9 and Figure 15). We favor the thermal pulse interpretation of the K-feldspar data for reasons described in section 5. The first ~150 Ma following peak Grenville metamorphism can be described by exponential cooling between ~650° and 300°C (Figure 15). At ~900 Ma, there is a significant decrease in cooling rate until ~750 Ma when rapid cooling takes place possibly due to a post orogenic tectonic adjustment [e.g., *Warnock and*

Table 5. $^{40}Ar/^{39}Ar$ Analytical Results: Tamarack Hornblende

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
900	10	158.93	0.3	0.2915	0.6	0.7824	0.9	0.20882	1.0	1.32e-14	0.3	0.080	61.1	97.289	881.91	8.42
920	10	96.99	0.3	0.2258	0.6	0.4033	1.5	0.00470	7.5	3.58e-15	0.3	0.102	98.2	95.617	869.89	7.18
950	10	99.84	0.3	0.2096	0.6	0.5871	0.9	0.00275	9.6	4.87e-15	0.3	0.132	98.9	99.069	894.63	7.33
970	10	103.53	0.3	0.1679	0.6	0.9081	1.6	0.00204	11.6	5.87e-15	0.3	0.167	99.2	103.010	922.45	7.49
990	10	106.77	0.3	0.1232	0.6	1.1581	0.8	0.00180	9.9	7.16e-15	0.3	0.211	99.3	106.350	945.75	7.61
1010	10	110.66	0.3	0.0816	0.6	1.4685	0.8	0.00076	17.6	9.62e-15	0.3	0.269	99.7	110.590	974.85	7.78
1030	10	113.36	0.3	0.0596	0.6	1.6310	0.8	0.00108	10.4	1.36e-14	0.3	0.352	99.6	113.220	992.68	7.92
1050	10	114.24	0.3	0.0484	0.6	1.7411	0.8	0.00111	9.1	1.37e-14	0.3	0.435	99.6	114.110	998.67	7.95
1070	10	114.97	0.3	0.0445	0.6	1.7957	0.8	0.00065	17.3	1.56e-14	0.3	0.531	99.8	114.980	1004.50	7.98
1090	10	114.45	0.3	0.0435	0.6	1.8034	0.8	0.00060	14.4	1.49e-14	0.3	0.621	99.8	114.480	1001.20	7.97
1110	10	113.81	0.3	0.0427	0.6	1.8230	0.8	0.00109	10.8	1.06e-14	0.3	0.686	99.6	113.700	995.90	7.93
1130	10	113.90	0.3	0.0438	0.7	1.7850	0.8	0.00052	23.3	1.08e-14	0.3	0.751	99.8	113.950	997.57	7.91
1150	10	115.13	0.3	0.0443	0.8	1.8060	0.8	-0.00009	100.0	1.03e-14	0.3	0.814	99.9	115.370	1007.10	7.97
1180	10	113.90	0.3	0.0436	0.6	1.8153	0.8	0.00107	6.6	1.75e-14	0.3	0.920	99.7	113.790	996.53	7.95
1210	10	113.09	0.3	0.0424	1.3	1.7416	0.9	0.00139	14.5	6.03e-15	0.3	0.957	99.4	112.870	990.31	7.89
1250	10	114.01	0.3	0.0412	1.3	1.7448	1.2	0.00194	31.4	2.80e-15	0.3	0.974	99.0	113.630	995.45	8.01
1350	10	114.48	0.3	0.0411	1.2	1.6852	0.8	0.00082	41.4	4.31e-15	0.3	1.000	99.5	114.430	1000.80	7.97

$J = (6.48 \pm 0.06)$ x10-3; $(^{36}Ar/^{37}Ar)_{Ca} = (6.91 \pm 0.69)$ x10-5; $(^{39}Ar/^{37}Ar)_{Ca} = (9.42 \pm 0.94)$ x10-4; $(^{40}Ar/^{39}Ar)_K = (2.62 \pm 0.26)$ x10-2; $(^{38}Ar/^{39}Ar)_K = (1.160 \pm 0.12)$ x10-2; mass discrimination is 1.0245 ± 0.01; weight is 0.0041 mg. Dur., duration; s.d., standard deviation.

*In percent.

Table 6. $^{40}Ar/^{39}Ar$ Analytical Results: 94GG01a Biotite

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
600	10	1624.10	0.4	1.1916	0.7	0.5580	4.5	5.24430	1.0	4.78e-16	0.4	0.002	4.6	74.470	710.40	120.02
700	10	132.35	0.3	0.1461	0.6	0.6633	2.7	0.10431	1.4	7.32e-16	0.3	0.005	75.9	101.580	912.41	8.09
800	10	105.46	0.3	0.1211	0.6	0.0583	1.3	0.01521	1.2	7.11e-15	0.3	0.035	95.6	100.950	907.95	7.40
900	10	100.76	0.3	0.1152	0.6	0.0224	1.2	0.00164	4.1	3.55e-14	0.3	0.186	99.5	100.250	903.01	7.35
950	10	100.83	0.3	0.1139	0.6	0.0191	0.8	0.00043	24.4	2.08e-14	0.3	0.274	99.8	100.680	906.05	7.38
980	10	99.95	0.3	0.1151	0.6	0.0328	5.6	0.00038	13.1	1.44e-14	0.3	0.335	99.8	99.815	899.94	7.37
1020	10	100.26	0.3	0.1243	0.6	0.0520	8.6	0.00067	11.4	1.48e-14	0.3	0.397	99.7	100.040	901.55	7.37
1060	10	100.46	0.3	0.1294	0.6	0.0744	0.8	0.00077	18.0	1.75e-14	0.3	0.471	99.7	100.220	902.79	7.36
1100	10	100.80	0.3	0.1282	0.6	0.0919	6.3	0.00166	4.8	3.08e-14	0.3	0.602	99.4	100.300	903.37	7.34
1150	10	100.18	0.3	0.1146	0.6	0.0267	10.5	0.00146	6.2	6.05e-14	0.3	0.859	99.5	99.722	899.28	7.33
1250	10	99.97	0.3	0.1131	0.6	0.0606	1.1	0.00209	6.9	3.33e-14	0.3	1.000	99.3	99.337	896.54	7.30

$J = (6.48 \pm 0.06)$ x10-3; $(^{36}Ar/^{37}Ar)_{Ca} = (6.91 \pm 0.69)$ x10-5; $(^{39}Ar/^{37}Ar)_{Ca} = (9.42 \pm 0.94)$ x10-4; $(^{40}Ar/^{39}Ar)_K = (2.62 \pm 0.26)$ x10-2; $(^{38}Ar/^{39}Ar)_K = (1.160 \pm 0.12)$ x10-2; mass discrimination is 1.0245 ± 0.01; weight is 0.0014 mg. Dur., duration; s.d., standard deviation.

*In percent.

Zeitler, 1998]. Between ~700 and ~600 Ma, the thermal history is constrained to a maximum temperature of 100°C prior to a ~1 m.y. thermal pulse at ~600 Ma.

5. Discussion

5.1. Age of Magnetizations

The Hb_A component solely resides in SD magnetite. In order to date the magnetization successfully, the magnetization must also be thermoremanent (TRM) in origin. The observation that the magnetite is symplectically intergrown with fine-grained orthopyroxene strongly supports a thermoremanent origin for the remanence since this texture is characteristic of a high-temperature origin [*Barker*, 1990]. A similar mineralogic association has been studied by *Barton and Van Gaans* [1988] in Norway. They were able to use pyroxene geothermometry to constrain the temperature of formation of the magnetite-orthopyroxene symplectites to between 600°C and 700°C.

The fact that LTD has a negligible effect on the Hb_A component, that the remanence is removed over a very narrow range in temperature, and that symplectic magnetite (0.5–10 μm) spanning the SD to MD size ranges hosted by orthopyroxene has been observed indicates that the Hb_A component is a TRM carried by SD magnetite. Therefore, using Néel theory to describe the magnetic relaxation of the remanence is reasonable. Because the unblocking temperature (560° ± 5°C) is near the Curie temperature (570°C), the correction for magnetic relaxation is small (5°C) and relatively independent of the cooling rate [*Dodson and McClelland-Brown*, 1980]. Therefore the magnetization was blocked at an in situ temperature of 555° ± 5°C. By comparing this temperature to the thermochronometric cooling curve in Figure 15, an age of 1015 ± 15 Ma for the Hb_A pole is obtained. This is older than the 980 Ma age assigned by *Berger et al.* [1979] largely due to our refinements to the cooling history using modern techniques.

Although we found a pyrrhotite direction that could be considered the Hb_C component, we believe that this is coincidental in light of the other, randomly distributed pyrrhotite directions found (Figure 11). Because *Buchan and Dunlop* [1976] report that the Hb_C direction is primarily carried by magnetite and because the locality in which it was found is only a short distance from the lobe and the primary Hb_A, we believe that the Hb_C

Table 7. $^{40}Ar/^{39}Ar$ Analytical Results: 94GG01b Biotite

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
700	10	457.21	0.3	0.4078	0.6	0.8250	2.0	1.23820	1.0	1.53e-15	0.3	0.008	19.9	91.370	838.97	27.47
800	10	101.84	0.3	0.1425	0.6	0.1531	4.0	0.01326	1.2	8.54e-15	0.3	0.053	96.0	97.912	886.37	7.24
850	10	99.66	0.3	0.1366	0.6	0.0383	8.7	0.00304	1.9	1.49e-14	0.3	0.132	99.0	98.742	892.30	7.34
900	10	99.78	0.3	0.1349	0.6	0.0430	1.4	0.00176	4.3	1.64e-14	0.3	0.218	99.4	99.237	895.83	7.33
950	10	99.50	0.3	0.1344	0.6	0.0148	4.6	0.00041	15.7	1.56e-14	0.3	0.301	99.8	99.351	896.64	7.29
980	10	99.51	0.3	0.1367	0.6	0.0316	3.9	0.00099	9.4	1.20e-14	0.3	0.365	99.6	99.194	895.52	7.31
1020	10	99.42	0.3	0.1515	0.6	0.0792	8.9	0.00053	36.5	1.18e-14	0.3	0.427	99.7	99.245	895.89	7.30
1060	10	99.34	0.3	0.1845	0.6	0.2394	2.3	0.00131	5.1	1.26e-14	0.3	0.494	99.5	98.958	893.84	7.29
1100	10	99.58	0.3	0.1830	0.6	0.2688	0.9	0.00169	2.5	1.90e-14	0.3	0.594	99.4	99.081	894.72	7.29
1125	10	97.58	0.3	0.1523	0.6	0.1532	1.1	0.00153	4.9	2.09e-14	0.3	0.705	99.4	97.122	880.72	7.19
1150	10	98.40	0.4	0.1340	0.6	0.0626	4.6	0.00112	2.9	2.64e-14	0.3	0.844	99.6	98.048	887.35	7.43
1250	10	99.51	0.3	0.1342	0.6	0.1163	5.0	0.00159	1.9	2.94e-14	0.3	1.000	99.4	99.023	894.31	7.29

$J = (6.48 \pm 0.06)$ x10-3; $(^{36}Ar/^{37}Ar)_{Ca} = (6.91 \pm 0.69)$ x10-5; $(^{39}Ar/^{37}Ar)_{Ca} = (9.42 \pm 0.94)$ x10-4; $(^{40}Ar/^{39}Ar)_K = (2.62 \pm 0.26)$ x10-2; $(^{38}Ar/^{39}Ar)_K = (1.160 \pm 0.12)$ x10-2; mass discrimination is 1.0245 ± 0.01; weight is 0.0014 mg. Dur., duration; s.d., standard deviation.

*In percent.

Table 8. $^{40}Ar/^{39}Ar$ Analytical Results: MG1 K-feldspar

Temp. °C	Dur. min	40/39	s.d.*	38/39	s.d.*	37/39	s.d.*	36/39	s.d.*	^{39}Ar mol	s.d.*	Fraction ^{39}Ar	$^{40}Ar^*$	$^{40}Ar^*/^{39}Ar_K$	Age Ma	s.d. Ma
425	10	#####	0.4	0.8054	1.1	0.3263	11.9	3.92200	1.1	1.74e-15	0.4	0.001	15.8	217.110	1584.60	57.24
450	10	482.46	0.4	0.2490	3.9	0.5873	5.3	1.31070	1.1	1.02e-15	0.4	0.001	19.6	95.177	866.71	29.06
450	20	485.98	0.3	0.2449	3.3	0.8495	4.9	1.36390	1.1	1.23e-15	0.3	0.002	17.0	83.020	776.59	31.94
480	10	552.02	0.3	0.2689	3.5	0.5447	16.9	1.47410	1.1	1.05e-15	0.3	0.002	21.0	116.450	1014.30	31.69
480	20	564.28	0.3	0.3139	1.7	0.4936	1.0	1.65280	1.0	1.93e-15	0.3	0.003	13.4	75.912	721.74	38.33
510	10	358.74	0.3	0.1816	3.1	0.3464	3.0	0.93090	1.0	1.83e-15	0.3	0.004	23.2	83.664	781.48	21.81
510	20	159.06	0.3	0.0644	4.5	0.1883	6.1	0.33384	1.3	3.37e-15	0.3	0.005	37.8	60.396	595.86	11.57
540	10	97.19	0.3	0.0284	12.2	0.1143	22.0	0.09184	1.2	2.76e-15	0.3	0.007	71.4	70.032	675.06	6.45
540	20	63.44	0.3	0.0102	17.4	0.1055	12.3	0.03018	1.5	5.42e-15	0.3	0.009	85.3	54.504	545.66	5.06
580	10	67.00	0.3	0.0205	7.2	0.1121	18.7	0.01326	2.4	6.53e-15	0.3	0.012	93.6	63.061	618.11	5.59
580	20	56.86	0.3	0.0073	13.5	0.0029	100.0	0.00463	5.1	9.88e-15	0.3	0.016	97.1	55.471	554.00	4.97
620	10	62.55	0.3	0.0129	6.6	0.0309	8.3	0.00226	6.7	1.14e-14	0.3	0.021	98.5	61.864	608.16	5.35
620	20	60.35	0.3	0.0098	5.0	0.0549	8.0	0.00230	5.6	2.00e-14	0.3	0.030	98.6	59.647	589.56	5.23
660	10	65.69	0.3	0.0135	3.7	0.0258	13.9	0.00246	8.5	2.00e-14	0.3	0.039	98.6	64.942	633.66	5.55
660	20	65.41	0.3	0.0101	3.2	0.0190	9.7	0.00153	3.7	3.06e-14	0.3	0.052	99.1	64.932	633.57	5.52
700	10	68.44	0.3	0.0112	3.0	0.0447	5.5	0.00108	7.2	2.90e-14	0.3	0.065	99.3	68.100	659.46	5.73
750	10	70.91	0.3	0.0095	6.5	0.0612	10.1	0.00021	100.0	1.59e-14	0.3	0.072	99.6	70.832	681.48	5.89
750	40	72.04	0.3	0.0127	1.6	0.0823	4.6	0.00209	1.9	1.06e-13	0.3	0.119	99.1	71.403	686.05	5.90
750	80	74.05	0.3	0.0118	1.3	0.0333	8.0	0.00155	4.3	1.08e-13	0.3	0.167	99.3	73.567	703.27	6.06
800	10	74.83	0.3	0.0088	5.4	0.0220	3.8	0.00021	45.9	2.05e-14	0.3	0.176	99.6	74.744	712.56	6.10
850	10	76.03	0.3	0.0111	1.9	0.0124	20.8	0.00093	4.7	4.96e-14	0.3	0.198	99.5	75.728	720.29	6.14
900	10	76.28	0.3	0.0116	1.3	0.0263	40.5	0.00110	2.2	7.79e-14	0.3	0.232	99.5	75.933	721.90	6.16
900	60	77.50	0.3	0.0119	0.8	0.0362	2.6	0.00108	1.5	1.44e-13	0.3	0.296	99.5	77.153	731.43	6.22
900	120	79.43	0.3	0.0137	1.1	0.1231	3.0	0.00326	3.0	7.24e-14	0.3	0.328	98.7	78.457	741.57	6.30
930	10	80.38	0.3	0.0040	30.7	0.0124	34.5	-0.00038	100.0	7.75e-15	0.3	0.331	99.3	80.461	757.03	6.50
930	120	81.85	0.3	0.0126	1.1	0.0466	4.0	0.00205	1.9	8.52e-14	0.3	0.369	99.2	81.225	762.89	6.43
980	60	82.25	0.3	0.0128	1.1	0.0309	9.0	0.00080	13.2	7.72e-14	0.3	0.403	99.6	81.989	768.74	6.47
1000	180	83.13	0.3	0.0134	0.7	0.0178	9.7	0.00144	2.1	1.88e-13	0.3	0.486	99.4	82.678	773.99	6.51
1050	10	82.69	0.3	0.0111	5.7	0.0106	48.1	0.00142	6.9	1.55e-14	0.3	0.493	99.0	82.248	770.71	6.51
1100	10	82.34	0.3	0.0157	1.6	0.0776	12.1	0.00049	35.1	4.36e-14	0.3	0.512	99.6	82.176	770.16	6.60
1150	10	82.67	0.3	0.0159	0.9	0.0000	100.0	0.00085	3.2	9.53e-14	0.3	0.554	99.6	82.396	771.84	6.51
1200	10	84.83	0.3	0.0179	0.7	0.0150	11.4	0.00072	13.6	1.98e-13	0.3	0.642	99.7	84.588	788.47	6.66
1300	10	86.09	0.3	0.0178	0.8	0.0195	9.2	0.00152	10.5	6.23e-13	0.3	0.917	99.4	85.612	796.19	6.67
1400	10	91.40	0.3	0.0159	1.0	0.0821	4.3	0.00118	26.0	1.72e-13	0.3	0.993	99.5	91.032	836.48	6.98
1450	10	93.25	0.3	0.0146	4.2	0.0145	32.4	0.00738	1.8	1.60e-14	0.3	1.000	96.7	91.044	836.58	6.97

$J = (6.48 \pm 0.06)$ x10-3; $(^{36}Ar/^{37}Ar)_{Ca} = (6.91 \pm 0.69)$ x10-5; $(^{39}Ar/^{37}Ar)_{Ca} = (9.42 \pm 0.94)$ x10-4; $(^{40}Ar/^{39}Ar)_K = (2.62 \pm 0.26)$ x10-2; $(^{38}Ar/^{39}Ar)_K = (1.160 \pm 0.12)$ x10-2; mass discrimination is 1.0269 ± 0.01; weight is 0.0019 mg. Dur., duration; s.d., standard deviation.

*In percent.

component is probably secondary. Until it is explored further, the Hb_C pole should not be used for plate reconstructions.

We were able to find the Hb_B component in samples which contain both magnetite and pyrrhotite and samples which contain only magnetite. Hb_B is always associated with the Hb_A component carried by SD magnetite. Where magnetite is the only magnetic phase, LTD removes the Hb_B component entirely (Figures 5 and 7), indicating that it is carried by magnetite exhibiting MD behavior. Where pyrrhotite coexists with magnetite, the Hb_B component is not completely removed until a demagnetization temperature of 320°C has been reached (Figure 6). The close association of the pyrrhotite and magnetite remanence could indicate that part of the magnetite resulted from the partial oxidation of pyrrhotite at high temperature [*Bina and Daly*, 1994]. However, we believe that it is unlikely that the significant number of samples that contain only pyrrhotite could have escaped such a thermal event. What is more difficult to explain is why the fine-grained rocks that contain only pyrrhotite record seemingly stable remanences (Figure 10) that are randomly distributed (Figure 11). One possibility is that these magnetizations could reside in fine-grained, SD pyrrhotite that is aligned by a strong magnetic anisotropy with the orientation of veins. We believe that these veins were emplaced during intrusion because of the lack of such veins in the coarse-grained rocks.

As discussed earlier in section 1, magnetite exhibiting MD behavior is not suitable for direct dating using its unblocking temperature, so we cannot assign an age to the Hb_B component in the same way we assigned the age to the Hb_A component. An alternative approach to dating a magnetization of this type is to try to correlate it with a low-temperature thermal event. The cooling history obtained from the K-feldspar can be interpreted in two ways: simple slow cooling or more rapid cooling followed by a thermal pulse of about 175°C at ~600 Ma (Figure 14d). Assuming simple slow cooling, one could interpret the age of the Hb_B pole to be ~900 Ma, since pyrrhotite passes through its Curie temperature at this point on the thermochronometric cooling curve (Figure 15). This is a discrete event for pyrrhotite; however, it is not for magnetite. Alternatively, the record of a mild, short-lived thermal event at ~600 Ma may not have been preserved in visible form in the rock record but may have been preserved only by the K-feldspar system and the magnetizations carried by MD magnetite and large MD pyrrhotite. Because magnetic blocking is more sensitive than isotopic closure to short thermal pulses, the magnetization can help distinguish between

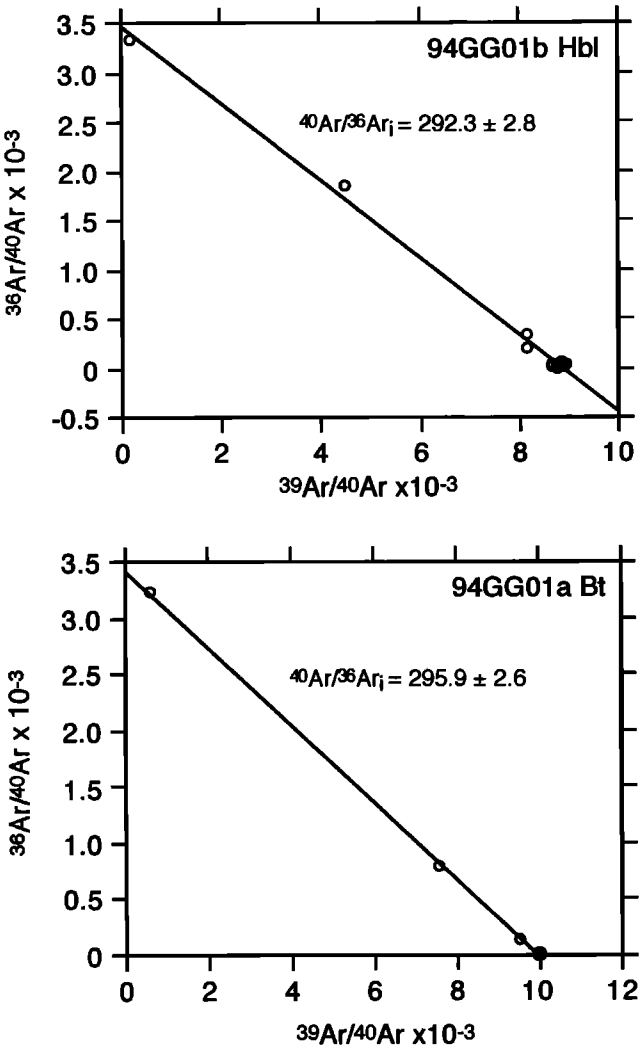


Figure 12. Inverse isochron plot of $^{40}Ar/^{39}Ar$ step-heating data from hornblende and biotite. Note that the trapped component (Y-intercept) is indistinguishable from atmospheric argon (295.5 [*Steiger and Jäger*, 1977]), indicating that there is no excess argon present.

the two K-feldspar models. The Hb_B pole is closer to the Cambrian poles than to the Hb_A pole (Figure 16), which suggests that the ~600 Ma thermal event model is most likely. The Long Range dikes [*Murthy et al.*, 1992] of Labrador have a U/Pb baddeleyite emplacement age of 615 ± 2 Ma [*Kamo and Gower*, 1994], indicating that at least part of the Grenville Province was tectonically active at this time.

5.2. Apparent Polar Wander

The Grenville path is partly composed of poles clustering around Hb_A which are interpreted as having been acquired following the peak Grenville metamorphism at ~1041 Ma [*Mezger et al.*, 1991] (Table 10 and Figure 16a). These poles are loosely defined as the "A poles" of the Grenville apparent polar wander path based on their proximity to the Hb_A pole and their "high" unblocking temperatures (typically 580° to 680°C). We have assigned an age of 1015 ± 15 Ma to the Hb_A pole, previously dated at 980 Ma. Since the Hb_A pole lies near the southern extent of the A poles, it also defines the southern extent of the combined Keweenawan/Grenville track (Figure 16a). The refined age for the end of this track produces a better Laurentia/Baltica fit with Congo/São Francisco but a poorer Laurentia/Baltica fit with the Kalahari plate in reconstructions of Rodinia [*Weil et al.*, 1998]. This is reasonable since the Congo/São Francisco path is more well defined than the Kalahari path.

The remaining poles from the Grenville Province define a broad distribution near the Cambrian end of the North American Phanerozoic APWP and are generally considered to be post-Grenvillian cooling overprints or "B poles" (Table 11 and Figure 16b). The "B pole" portion of the Grenville APWP is not well defined since the majority of poles from intrusives of the Grenville Province in Canada have been assigned ages greater

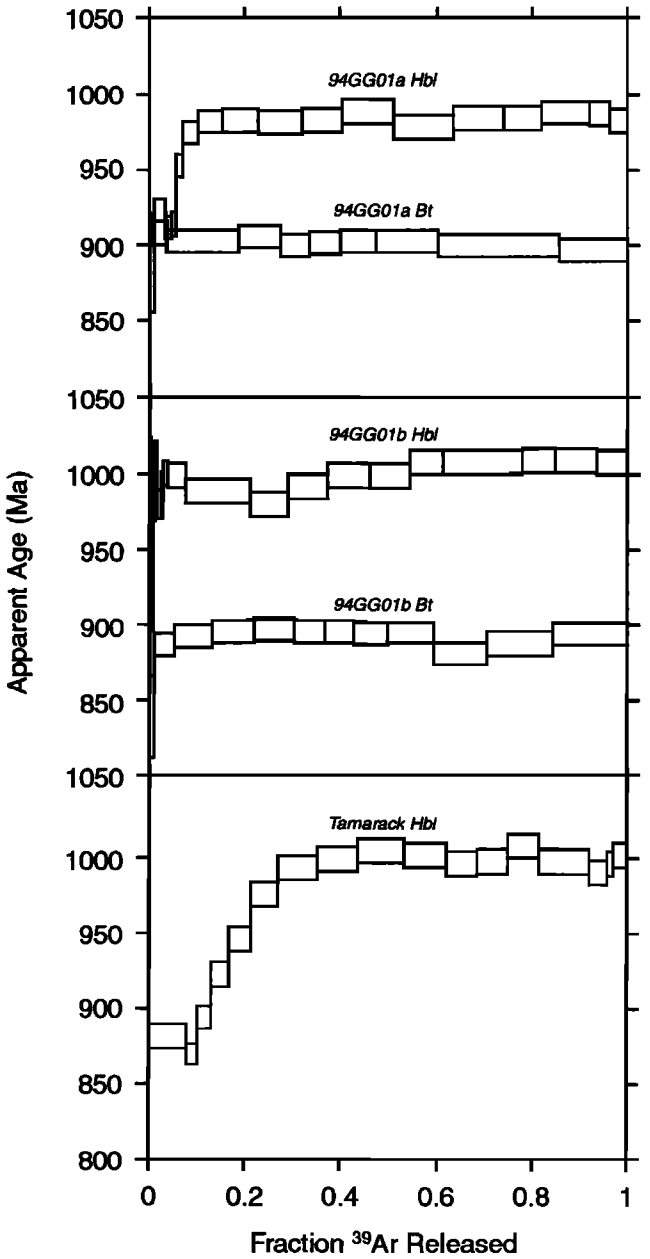


Figure 13. Hornblende and biotite age spectra for samples from the Glamorgan unit.

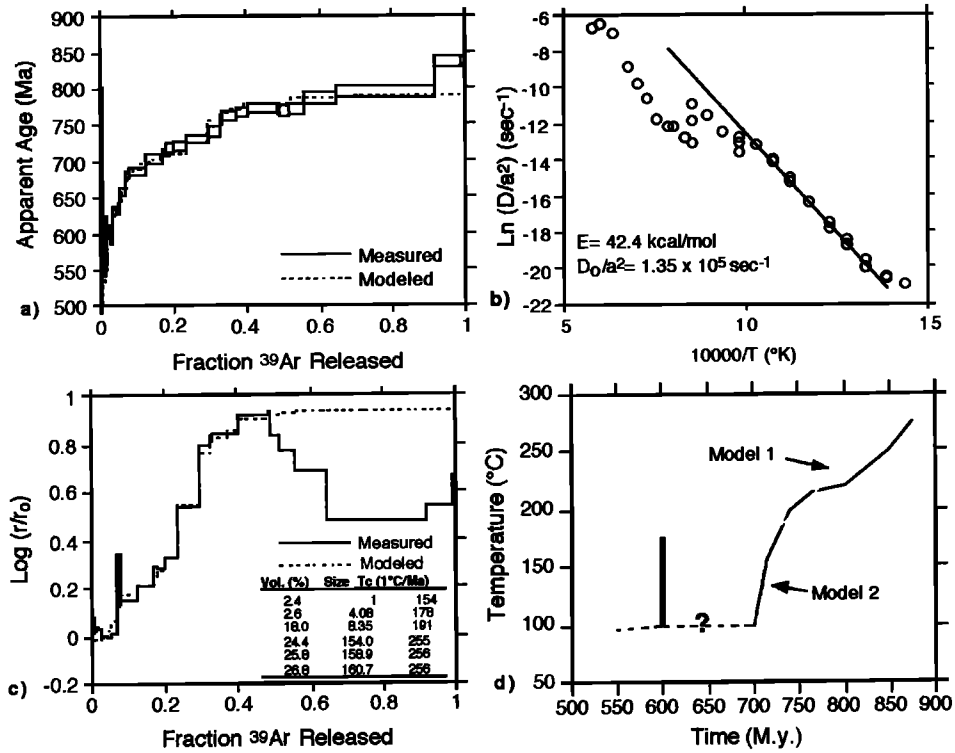


Figure 14. Results of detailed step-heating of K-feldspar sample 94MG01. (a) Measured and modeled age spectra. (b) Arrhenius plot of ^{39}Ar release data used to determine the activation energy and diffusion coefficient. Departure from linearity at higher temperatures is indicative of a range in diffusion domain volumes. (c) Log(r/r_O) plot of Arrhenius data used to model the relative size and volume percent of each diffusion domain. (d) Two possible thermal histories which can produce the observed age spectrum. Note that these are end-member solutions: one involving just linear slow cooling, and another with cooling followed by a ~1 m.y. thermal pulse. Note that the isothermal segment between 700 and 600 Ma is constrained to a maximum temperature of 100°C.

Table 9. Summary of Thermochronometric Data From the Bancroft Terrane

Mineral	Method	Sample Name	Age, Ma	1σ, Ma	Reference
Titanite	U/Pb	90-40	1045	1	*Mezger et al.*[1991]
Titanite	U/Pb	GO4b	1044	0.5	*Mezger et al.*[1991]
Titanite	U/Pb	5D	1046	0.5	*Mezger et al.*[1991]
Titanite	U/Pb	90-45	1034	1.5	*Mezger et al.*[1991]
Titanite	U/Pb	90-45	1032	1	*Mezger et al.*[1991]
Titanite	U/Pb	REN-28	1041	1	*Mezger et al.*[1991]
Titanite	U/Pb	BA-28	1045	1	*Mezger et al.*[1991]
Titanite	U/Pb	89-06/1	1043	1	*Mezger et al.*[1991]
Mean titanite*			1041	1	
Hornblende*	$^{40}Ar/^{39}Ar$	G21-113	971	8	*Berger and York*[1981]
Hornblende*	$^{40}Ar/^{39}Ar$	G16-25	998	7	*Berger and York*[1981]
Hornblende	$^{40}Ar/^{39}Ar$	Ban34	959	3†	*Cosca et al.*[1995]
Hornblende	$^{40}Ar/^{39}Ar$	Ban36	980	3†	*Cosca et al.*[1995]
Hornblende	$^{40}Ar/^{39}Ar$	Ban39	995	2†	*Cosca et al.*[1995]
Hornblende	$^{40}Ar/^{39}Ar$	Ban42	981	3†	*Cosca et al.*[1995]
Hornblende	$^{40}Ar/^{39}Ar$	Ban44	996	2†	*Cosca et al.*[1995]
Hornblende*	$^{40}Ar/^{39}Ar$	94GG01a	983	8	this study
Hornblende*	$^{40}Ar/^{39}Ar$	94GG01b	998	8	this study
Hornblende*	$^{40}Ar/^{39}Ar$	Tamarack	996	8	this study
Biotite*	$^{40}Ar/^{39}Ar$	G4-22	906	6	*Berger and York*[1981]
Biotite*	$^{40}Ar/^{39}Ar$	94GG01a	901	7	this study
Biotite*	$^{40}Ar/^{39}Ar$	94GG01b	891	7	this study
Phlogopite	$^{40}Ar/^{39}Ar$	GO-1B	893	3†	*Cosca et al.*[1992]
Phlogopite	$^{40}Ar/^{39}Ar$	WI-1B	904	3†	*Cosca et al.*[1992]
K-feldspar*	$^{40}Ar/^{39}Ar$	MG-1	see Figure 14		this study

* Ages used in Figure 15.
† Does not include error in J.

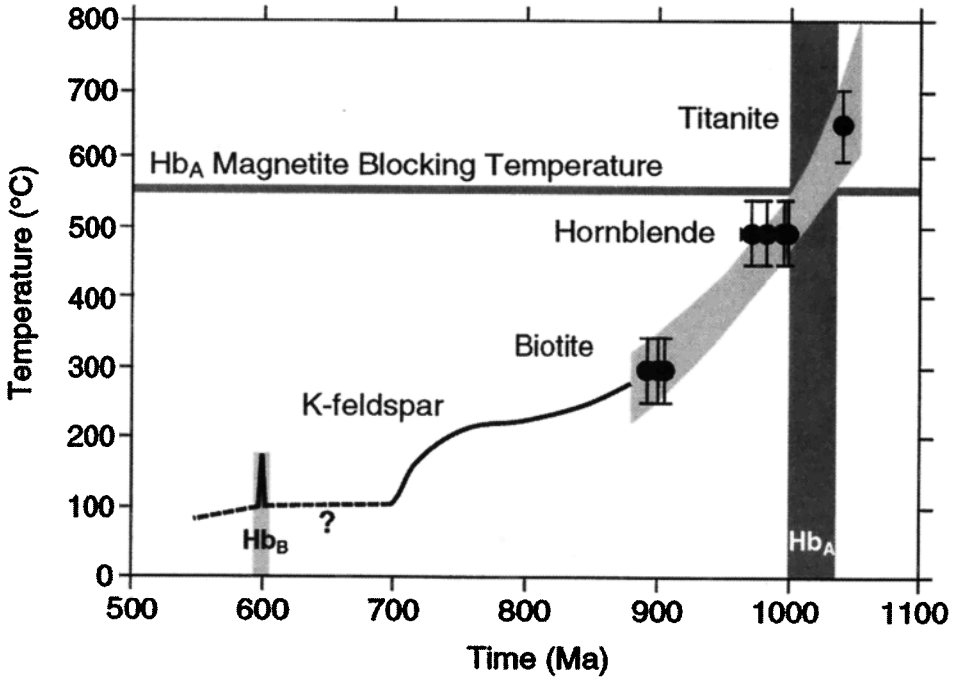


Figure 15. Thermochronologic cooling curve for the Glamorgan gabbro. The hornblende, biotite, and K-feldspar are from the lobe locality (Table 9). The metamorphic titanite age is a regional average (Table 9). Monte Carlo simulation was used to determine of the range of exponential curves that can be produced given the errors on both the ages and closure temperatures. The age of remanence of the Hb_A component is defined by the intersection of the magnetic blocking temperature (555° ± 5°C) and the simulated curves. This yields a magnetization age of 1015 ± 15 Ma for the Hb_A paleomagnetic pole. The Hb_B pole was likely reset during a thermal event at ~600 Ma.

than ~820 Ma, the original age assigned to Hb_B (Table 11). Of the 18 poles for this portion of the apparent polar wander path, 8 poles have been assigned ages based on their proximity to 10 "well-dated" poles, 7 of which have ages based on single mica or whole rock K/Ar cooling ages (Table 11). These ages typically reflect the time the unit passed through ~300°C and may have no relation to the magnetic blocking temperature. It is quite possible that many B poles were affected by the 600 Ma thermal pulse that remagnetized the Hb_B pole. If this assumption is correct, then there may be strong evidence for Laurentia's paleopole be-

Table 10. Grenville "A Poles" and Assigned Ages

Rock Unit	Latitude, deg	Longitude, deg	Age, Ma	Method	Reference
Poles with no assigned ages					
French River anorthosite	26S	135E	N/A		*Stupavsky and Symons* [1982]
Ottawa basic intrusions	32S	155E	N/A		*Irving et al.* [1972]
Poles with whole rock or mica ages					
Allard Lake anorthosite	39S	140E	<1025 ± 60	K/Ar - Biotite	*Hargraves and Burt* [1967]
Borden dikes	26.6S	153.3E	819 ± 107	K/Ar - WR	*Christie and Fahrig* [1983]
Morin anorthosite	42S	139E	<1124 ± 27	Rb/Sr - WR	*Irving et al.* [1974]
Magnetawan metasediments	24S	130E	~940	K/Ar - Mica	*McWilliams and Dunlop* [1975]
Mealy Mountain anorthosite	38S	178E	1500-1600	K/Ar	*Fahrig et al.* [1974]
Mealy Mountain anorthosite (NW)	8.5N	181E	1500-1600	K/Ar	*Fahrig et al.* [1974]
Mealy Mountain dikes (A)	22.9S	173.4W	1078-1222	K/Ar - WR	*Park and Emslie* [1983]
Mealy Mountain dikes (B)	34.2S	147.9E	1078-1222	K/Ar - WR	*Park and Emslie* [1983]
Poles with Ar/Ar ages without LTD					
Mattawa Tonalitic gneiss	21S	127E	1009 ± 2	Ar/Ar - Hbl	*Hyodo and Dunlop* [1993]
Thanet gabbro	30S	165E	1202 ± 10	Ar/Ar	*Buchan* [1978]
Tudor gabbro	17N	137E	1110 ± 20	Ar/Ar	*Palmer and Carmichael* [1973]
Tudor gabbro	12N	133E	1110 ± 20	Ar/Ar	*Dunlop et al.* [1985]
Whitestone quartz diorite	22S	146E	980-1010	Ar/Ar - CC	*Ueno et al.* [1975]
Poles with Ar/Ar and U/Pb ages with LTD					
Haliburton intrusions (Hb_A)	32.6S	141.9E	1015 ± 15	Ar/Ar - U/Pb	*Buchan and Dunlop* [1976], this study

WR, whole rock; CC, cooling curve; Hbl, hornbelnde. (Table modified from *Hyodo and Dunlop* [1993].)

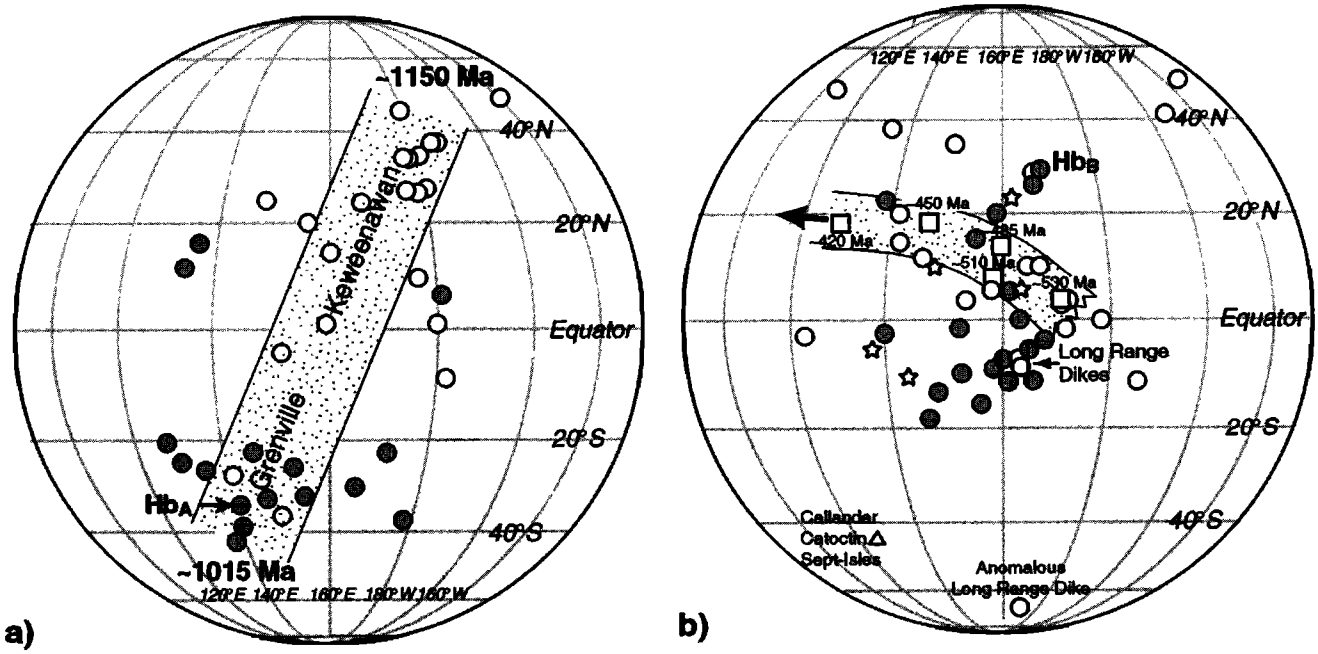


Figure 16. (a) Summary of Keweenawan poles (open circles) from *Weil et al.* [1998] and Grenville "A poles" (shaded circles) from Table 10 showing widely scattered trend from ~40°N, 180°E, ~1150 Ma down to ~40°S, 140°E, ~1015 Ma. Typical confidence limits vary from 5° to 15°. (b) Comparison of Grenville "B poles" (shaded circles) from Table 11 to other Late Precambrian poles: Stars are well-dated mean poles (plotted as north poles) ranging from 830 to 725 Ma from *Powell et al.* [1993]; open circles are Late Proterozoic to Ordovician poles from Table 6 of *Hyodo and Dunlop* [1993]; triangle is the high-latitude mean of the Callander Complex, Catoctin Formation, and Sept Isles dike from *Meert et al.* [1994]; squares define the Lower Paleozoic apparent polar wander path (shaded track) for North America from *Van der Voo* [1990]. Typical confidence limits vary from 5° to 15°.

Table 11. Grenville "B Poles" and Assigned Ages

Rock Unit	Latitude, deg	Longitude, deg	Age, Ma	Method	Reference
Poles with no assigned ages					
Cordova gabbro	24N	165E	N/A		*Dunlop and Stirling* [1985]
French River anorthosite	13N	154E	N/A		*Stupavsky and Symons* [1982]
Grenville Front anorthosite	8N	161E	N/A		*Palmer and Carmichael* [1973]
Lac St-Jean anorthosite (N)	8S	167E	N/A		*Buchan et al.* [1983]
Lac St-Jean anorthosite (R)	17S	147E	N/A		*Buchan et al.* [1983]
Thanet gabbro	20N	159E	N/A		*Buchan* [1978]
Wilberforce pyroxenite	14.5S	148E	N/A		*Palmer and Carmichael* [1973]
Whitestone anorthosite	16S	156E	N/A		*Ueno et al.* [1975]
Poles with K/Ar whole rock or mica ages					
Frontenac Axis dikes	12S	162E	817 ± 70	K/Ar - Biot	*Park and Irving* [1972]
Indian Head anorthosite	9.5S	158.5E	830 ± 42	K/Ar - Biot	*Murthy and Rao* [1975]
Morin anorthosite	0N	164E	910	K/Ar - Biot	*Irving et al.* [1974]
Steel Mountain anorthosite	22.5N	138E	451 ± 35	K/Ar - Chlor	*Murthy and Rao* [1975]
St-Urbain anorthosite	2S	154E	~890	K/Ar - Biot	*Robertson and Roy* [1979]
Umfraville gabbro	11S	166E	911 ± 11	K/Ar - WR	*Symons* [1978]
Umfraville gabbro	7.5S	160E	911 ± 11	K/Ar - WR	*Palmer et al.* [1979]
Poles with Ar/Ar ages without LTD					
Cordova gabbro	10.5S	151E	1052-1143	Ar/Ar - CC	*Dunlop and Stirling* [1985]
Mattawa Tonalitic gneiss	2S	140E	1009 ± 2	Ar/Ar - Hbl	*Hyodo and Dunlop* [1993]
Whitestone massive anorthosite	5S	168E	980-1010	Ar/Ar - CC	*Ueno et al.* [1975]
Poles with Ar/Ar and U/Pb ages with LTD					
Haliburton intrusions (Hb_B)	29.9N	169.9E	~600	Ar/Ar - CC	*Buchan and Dunlop* [1976], this study

Biot, biotite; Chlor, chlorite; Hbl, hornblende; WR, whole rock; CC, cooling curve. (Table modified from *Hyodo and Dunlop* [1993].)

ing near the equator just prior to the Cambrian. In order to test this hypothesis the LTD method must be applied to more B poles. The Long Range dikes have an age of magnetization bracketed by a U/Pb baddeleyite emplacement age of 615 ± 2 Ma [*Kamo and Gower*, 1994] and a K/Ar biotite cooling age of 553 ± 22 Ma [*Wanless et al.*, 1970] and a pole position which is not significantly different from the positions of the Grenville B poles (Figure 16b). This would suggest that the B poles are in fact ~600 m.y. old. However, an anomalous high-latitude pole obtained from three sites of the Long Range dikes has also been used in conjunction with poles from the Callander Complex, Catoctin Formation, and the Sept-Isles dike in a recent pole path model placing Laurentia at the south pole at 615 Ma [*Meert et al.*, 1994]. The application of the LTD method could help clarify these conflicting models.

If the Grenville B poles are indeed younger than previously thought, they may be associated with a standstill in apparent polar wander. *Powell et al.* [1993] recently summarized high-quality poles that range in age from 725 to 830 Ma. These poles, when interpreted as north poles, also cluster around the Cambrian end of the Phanerozoic APWP (Figure 16b). Therefore the Proterozoic pole path for Laurentia may have the following characteristics: a Keweenawan track starting near 40°N, 180°E at ~1150 Ma that trends toward the equator at ~160°E. This trend continues south recorded by Grenville A poles including the Hb_A pole near 40°S, 140°E at ~1015 Ma. Finally, the path returns to the equator near 160°E by ~830 Ma, where it remains until the Cambrian.

6. Conclusions

Low-temperature demagnetization used in conjunction with K-feldspar thermochronometry can provide a powerful way to date Precambrian multicomponent magnetizations. We have demonstrated this technique by refining the age of magnetization for two important Grenville paleomagnetic poles. The Hb_A pole was found to be carried by SD magnetite which exhibits a high-temperature crystal morphology. By relating its laboratory unblocking temperature, corrected for magnetic relaxation, to the local cooling history, we obtain an age of 1015 ± 15 Ma. The Hb_B component is found superimposed on the Hb_A component and is carried by both MD magnetite and pyrrhotite. A mild (~175°C) thermal event at ~600 Ma is probably responsible for remagnetizing these minerals long after Grenville cooling was complete. Unequivocal evidence of the Hb_C pole was not found in the rocks studied here. Competing models for Late Precambrian apparent polar wander should be tested by applying the techniques outlined here to other critical poles.

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References

Barker, A. J., *Introduction to Metamorphic Textures and Microstructures*, 162 pp., Blackie, New York, 1990.

Barton, M., and C. Van Gaans, Formation of orthopyroxene-Fe-Ti oxide symplectites in Precambrian intrusives, Rogaland, southwestern Norway, *Am. Mineral.*, *73*, 1046-1059, 1988.

Berger, G. W., and D. York, Geothermometry from $^{40}Ar/^{39}Ar$ dating experiments, *Geochim. Cosmochim. Acta*, *45*, 795-811, 1981.

Berger, G. W., D. York, and D. J. Dunlop, Calibration of Grenvillian paleopoles by $^{40}Ar/^{39}Ar$ dating, *Nature*, *277*, 46-48, 1979.

Bina, M., and L. Daly, Mineralogical change and self-reversed magnetizations in pyrrhotite resulting from partial oxidation: geophysical implications, *Phys. Earth Planet. Inter.*, *85*, 83-99, 1994.

Borradaile, G. J., Low-temperature demagnetization and ice-pressure demagnetization in magnetite and haematite, *Geophys. J. Int.*, *116*, 571-584, 1994.

Buchan, K. L., Magnetic overprinting in the Thanet gabbro complex, Ontario, *Can. J. Earth Sci.*, *15*, 1407-1421, 1978.

Buchan, K. L., and D. J. Dunlop, Paleomagnetism of the Haliburton intrusions: Superimposed magnetizations, metamorphism, and tectonics in the Late Precambrian, *J. Geophys. Res.*, *81*, 2951-2967, 1976.

Buchan, K. L., G. W. Berger, M. O. McWilliams, D. York, and D. J. Dunlop, Thermal overprinting of natural remanent magnetization and K/Ar ages in metamorphic rocks, *J. Geomagn. Geoelectr.*, *29*, 401-410, 1977.

Buchan, K. L., W. F. Fahrig, G. N. Freda, and R. A. Frith, Paleomagnetism of the Lac St-Jean anorthosite and related rocks, Grenville Province, Quebec, *Can. J. Earth Sci.*, *20*, 246-258, 1983.

Christie, K.W., and W. F. Fahrig, Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville loop, *Can. J. Earth Sci.*, *20*, 275-289, 1983.

Cosca, M. A., E. J. Essene, M. J. Kunk, and J. F. Sutter, Differential unroofing within the Central Metasedimentary Belt of the Grenville Orogen: Constraints from $^{40}Ar/^{39}Ar$ thermochronology, *Contrib. Mineral. Petrol.*, *110*, 211-225, 1992.

Cosca, M. A., E. J. Essene, K. Mezger, and B. A. van der Pluijm, Constraints on the duration of tectonic processes: Protracted extension and deep-crustal rotation in the Grenville orogen, *Geology*, *23*, 361-364, 1995.

Dickin, A. P., R. H. McNutt, and F. Marcantonio, Proterozoic crustal accretion on the southeastern Laurentian margin: Nd model age evidence from the Grenville Province in Ontario, Canada, *Geol. Soc. Am. Abstrs. Programs.*, *20*, 103, 1988.

Dodson, M. H., and E. McClelland-Brown, Magnetic blocking temperatures of single-domain grains during slow cooling, *J. Geophys. Res.*, *85*, 2625-2637, 1980.

Downing, B. W., Petrography, deformation and emplacement of the Glamorgan gabbro complex, Haliburton County, Ontario, M.S. thesis, 160 pp., Univ. of Toronto, Ont., 1973.

Dunlop, D. J., and K. S. Argyle, Separating multidomain and single-domain-like remanences in pseudo-single-domain magnetites (215-540 nm) by low-temperature demagnetization, *J. Geophys. Res.*, *96*, 2007-2017, 1991.

Dunlop, D. J., and Ö. Özdemir, Thermal demagnetization of VRM and pTRM of single domain magnetite: No evidence for anomalously high unblocking temperatures, *Geophys. Res. Lett.*, *20*, 1939-1942, 1993.

Dunlop, D. J. and J. M. Stirling, Post-tectonic magnetization from the Cordova gabbro, Ontario and Paleozoic reactivation in the Grenville Province, *Geophys. J. R. Astron. Soc.*, *81*, 521-550, 1985.

Dunlop, D. J., and S. Xu, Theory of partial thermoremanent magnetization in multidomain grains, 1, Repeated identical barriers to wall motion (single microcoercivity), *J. Geophys. Res.*, *99*, 9005-9023, 1994.

Dunlop, D. J., H. Hyodo, T. Knight, and A. G. Steele, Paleomagnetism of the Tudor gabbro, Ontario: Evidence for divergence between Grenvillian and interior Laurentia, *Geophys. J. R. Astron. Soc.*, *83*, 699-720, 1985.

Fahrig, W. F., K. W. Christie, and E. J. Schwarz, Paleomagnetism of the Mealy Mountain anorthosite suite and of the Shabogamo gabbro, Labrador, Canada, *Can. J. Earth Sci.*, *11*, 18-29, 1974.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295-305, 1953.

Halgedahl, S. L., Experiments to investigate the origin of anomalously elevated unblocking temperatures, *J. Geophys. Res.*, *98*, 22,443-22,460, 1993.

Halgedahl, S. L., and M. Fuller, The dependence of magnetic domain structure upon magnetization state with emphasis upon nucleation as a mechanism for pseudo-single-domain behavior, *J. Geophys. Res.*, *88*, 6505-6522, 1983.

Hargraves, R. B., and D. M. Burt, Paleomagnetism of the Allard Lake anorthosite suite, *Can. J. Earth Sci.*, *4*, 357-369, 1967.

Harrison T. M., M. T. Heizler and O. M. Lovera, In vacuo crushing experiments and K-feldspar thermochronometry, *Earth Planet. Sci. Lett., 117*, 169-180, 1993.

Heaman, L. M., D. W. Davis, T. E. Krogh, and S. B. Lumbers, Geological evolution of the Central Metasedimentary Belt (CMB), Ontario: A U-Pb perspective, *Can. Geophys. Union Joint Annu. Meet., 12*, 54, 1987.

Hoffman, P. F., Precambrian geology and tectonic history of North America, in *The Geology of North America*, vol. A, edited by A. W. Bally and A. R. Palmer, pp. 447-511, Geol. Soc. Of Am., Boulder, Colo., 1989.

Hyodo, H., and D. J. Dunlop, Multicomponent magnetizations in Nipissing diabase near Temagami, Ontario, and structural and thermal history of the Grenville Front, *Can. J. Earth Sci., 26*, 467-478, 1989.

Hyodo, H., and D. J. Dunlop, Effect of anisotropy on the paleomagnetic contact test for a Grenville dike, *J. Geophys. Res., 98*, 7997-8017, 1993.

Irving, E., J. K. Park, and J. L. Roy, Paleomagnetism and the origin of the Grenville Front, *Nature, 236*, 344-346, 1972.

Irving, E., J. K. Park, and R. F. Emslie, Paleomagnetism of the Morin complex, *J. Geophys. Res., 79*, 5482-5490, 1974.

Jackson, M., and R. Van der Voo, Thermally activated viscous remanence in some magnetite- and hematite-bearing dolomites, *Geophys. Res. Lett., 13*, 1434-1437, 1986.

Kamo, S. L., and C. F. Gower, U-Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Ranges dykes, southeastern Labrador, *Atl. Geol., 30*, 259-262, 1994.

Kent, D. V., Thermoviscous remagnetization in some Appalachian limestones, *Geophys. Res. Lett., 12*, 805-808, 1985.

Kirschvink, J., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Kobayashi, K., and M. Fuller, Stable remanence and memory of multidomain materials with special reference to magnetite, *Philos. Mag., 18*, 601-624, 1968.

Lovera O. M., F. M. Richter, and T. M. Harrison, The $^{40}Ar/^{39}Ar$ thermochronology for slowly cooled samples having a distribution of diffusion domain sizes, *J. Geophys. Res., 94*, 17,917-17,931, 1989.

Lovera O. M., F. M. Richter, and T. M. Harrison, Diffusion domains determined by ^{39}Ar released during step heating, *J. Geophys. Res., 96*, 2057-2070, 1991.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett., 17*, 159-162, 1990.

McDougall, I., and T. M. Harrison, *Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method, Oxford Monogr. Geol. Geophys.*, vol. 9, 212 pp., Oxford Univ. Press, New York, 1988.

McFadden, P. L., and F. J. Lowes, The discrimination of mean directions drawn from Fisher distributions, *Geophys. J. R. Astron. Soc., 67*, 19-33, 1981.

McWilliams, M. O., and D. J. Dunlop, Precambrian paleomagnetism: Magnetization reset by the Grenville orogeny, *Science, 190*, 269-272, 1975.

Meert, J. G., R. Van der Voo, and T. W. Payne, Paleomagnetism of the Catoctin volcanic province: A new Vendian-Cambrian apparent polar wander path for North America, *J. Geophys. Res., 99*, 4625-4641, 1994.

Merrill, R. T., Low-temperature treatment of magnetite and magnetite-bearing rocks, *J. Geophys. Res., 75*, 3343-3349, 1970.

Mezger, K., B. A. van der Pluijm, E. J. Essene, and A. N. Halliday, Synorogenic collapse: A perspective from the middle crust, the Proterozoic Grenville Orogen, *Science, 254*, 695-698, 1991.

Middleton, M. F., and P. W. Schmidt, Paleothermometry of the Sydney Basin, *J. Geophys. Res., 87*, 5351-5359, 1982.

Murthy, G. S., and K. V. Rao, Paleomagnetism of Steel Mountain and Indian Head anorthosites from western Newfoundland, *Can. J. Earth Sci., 13*, 75-83, 1975.

Murthy, G., C. Gower, M. Tubrett, and R. Pätzold, Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador, Canada, *Can. J. Earth Sci., 29*, 1224-1234, 1992.

Palmer, H. C., and C. M. Carmichael, Paleomagnetism of some Grenville Province rocks, *Can. J. Earth Sci., 10*, 1175-1189, 1973.

Palmer, H. C., A. Hayatsu, C. E. Waboso, and S. Pullan, A paleomagnetic and K-Ar study of the Umfraville gabbro, Ontario, *Can. J. Earth Sci., 16*, 459-471, 1979.

Park, J. K., and R. F. Emslie, Paleomagnetic history of the Mealy dykes of Labrador, Canada, *Can. J. Earth Sci., 20*, 1818-1833, 1983.

Park, J. K., and E. Irving, Magnetism of dikes of the Frontenac Axis, *Can. J. Earth Sci., 9*, 763-765, 1972.

Powell, C. McA., Z. X. Li, M. W. McElhinny, J. G. Meert, and J. K. Park, Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana, *Geology, 21*, 889-892, 1993.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett., 28*, 133-143, 1975.

Robertson, P. B., and J. L. Roy, Shock diminished paleomagnetic remanence at the Charlevoix impact structure Quebec, *Can. J. Earth Sci., 16*, 1842-1856, 1979.

Samson, S. D., and E. C. Alexander Jr., Calibration of the interlaboratory $^{40}Ar/^{39}Ar$ dating standard, MMhb-1, *Chem. Geol., 66*, 27-34, 1987.

Soffel, H. C., Pseudo-single-domain effects and single-domain multidomain transition in natural pyrrhotite deduced from domain structure observations, *J. Geophys., 42*, 351-359, 1977.

Steiger R. H., and E. Jäger, Convention on the use of decay constants in geo- and cosmochronology, *Earth Planet. Sci. Lett., 36*, 359-362, 1977.

Stupavsky, M., and D. T. A. Symons, Isolation of early Paleohelikian remanence in Grenville anorthosites of the French River area, Ontario, *Can. J. Earth Sci., 19*, 819-828, 1982.

Symons, D. T. A., Paleomagnetism of the 1180 Ma Grenvillian Umfraville gabbro, Ontario, *Can. J. Earth Sci., 15*, 956-962, 1978.

Ueno, H., E. Irving, and R. H. McNutt, Paleomagnetism of the Whitestone anorthosite and diorite, the Grenville polar track, and relative motions of the Laurentian and Baltic Shields, *Can. J. Earth Sci., 12*, 209-226, 1975.

Van der Voo, R., Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, *Rev. Geophys., 28*, 167-206, 1990.

Wanless, R. K., R. D. Stevens, G. R. Lachance, and R. N. Delabio, Age determinations and geologic studies, *Geol. Surv. Can. Pap. 69-2A*, 1970.

Warnock, A. C., and P. K. Zeitler, $^{40}Ar/^{39}Ar$ thermochronometry of K-feldspar from the KTB borehole, Germany, *Earth Planet. Sci. Lett., 158*, 67-79, 1998.

Watson, G. S., A test for randomness of directions, *Mon. Not. Geophys. J. R. Astron. Soc., 7*, 160-161, 1956.

Weil, A. B., R. Van der Voo, C. M. Niocaill, and J. G. Meert, The Proterozoic supercontinent Rodinia: Paleomagnetically derived reconstructions for 1,100 to 800 Ma, *Earth Planet. Sci. Lett., 154*, 13-24, 1998.

Xu, S., and D. J. Dunlop, Theory of partial thermoremanent magnetization in multidomain grains, 2, Effect of microcoercivity distribution and comparison with experiment, *J. Geophys. Res., 99*, 9025-9033, 1994.

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